CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

DD SILVEIRA & SON DAIRY #2 (PREVIOUSLY TRIPLE J CREEK DAIRY)

22484 W CHINA CAMP RD

LOS BANOS CA 93635

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: DD Silveira & Son Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
22484 W China Camp RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1981

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X170-X013-XXXX

B. OPERATORS

Nunes, Jose & Odette

Operator name: Nunes, Jose & Odette

Telephone no.: (209) 826-1817 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11132 Wilson Rd RD	Los Banos	CA	93635

Silveira, Robert

Operator name: Silveira, Robert

Telephone no.: (Landline) (209) 604-1517 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2350	Los Banos	CA	93635

This operator is responsible for paying permit fees.

C. OWNERS

Nunes, Jose & Odette

Legal owner name: Nunes, Jose & Odette

Telephone no.: (209) 826-1817 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11132 Wilson Rd RD	Los Banos	CA	93635

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	83	0	0	30	0
Number under roof	814	72	518	259	97	93
Maximum number	814	155	518	259	127	93
Average number	814	155	518	259	127	93
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 32,040.06 *tons per reporting period*

Total nitrogen from manure: 386,266.20 *lbs per reporting period*

After ammonia losses (30% loss applied): 270,386.34 *lbs per reporting period*

Total phosphorus from manure: 63,224.70 *lbs per reporting period*

Total potassium from manure: 155,888.51 *lbs per reporting period*

Total salt from manure: 418,914.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,878,000 *gallons*

Total nitrogen generated: 62,438.60 *lbs*

Total phosphorus generated: 7,016.11 *lbs*

Total potassium generated: 114,995.77 *lbs*

Total salt generated: 609,446.03 *lbs*

	10,878,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,878,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Freshwater Pond	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/28/2023	Solid commercial fertilizer *UN-32*	23.70 *ton*	Dry-weight	1.0	99.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	46,456.74	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	46,456.74	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
N1	16	16	0	none	X070-X170-X014-XXXX
N10	16	16	1	manure	X070-X190-X008-XXXX
N11	68	68	2	manure	X070-X180-X007-XXXX X070-X180-X056-XXXX X070-X180-X057-XXXX X070-X180-X058-XXXX
N2	31	31	1	none	X070-X170-X012-XXXX
N3	102	102	1	process wastewater	X070-X170-X012-XXXX X070-X170-X023-XXXX X070-X170-X024-XXXX X070-X170-X025-XXXX
N4 (Picardo)	39	39	1	manure	X070-X170-X007-XXXX
N5	62	62	1	manure	X070-X190-X003-XXXX X070-X190-X005-XXXX
N6	37	37	1	manure	X070-X190-X003-XXXX X070-X190-X005-XXXX
N7	20	20	1	manure	X070-X190-X006-XXXX
N8	15	15	1	manure	X070-X190-X006-XXXX
N9	23	23	1	manure	X070-X190-X008-XXXX
W1	61	61	2	manure	X070-X150-X010-XXXX X070-X190-X001-XXXX
W13	82	82	2	manure	X070-X140-X030-XXXX
W14	87	87	2	manure	X070-X140-X030-XXXX
W15	48	48	2	process wastewater	X070-X140-X031-XXXX X070-X140-X032-XXXX
W16	75	75	2	process wastewater	X070-X140-X031-XXXX X070-X140-X032-XXXX X070-X140-X034-XXXX X070-X140-X035-XXXX
W2	60	60	2	manure	X070-X150-X010-XXXX X070-X190-X001-XXXX
W20	39	39	1	process wastewater	X070-X170-X031-XXXX
W23	73	73	2	manure	X070-X170-X031-XXXX X070-X170-X032-XXXX

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
W3	30	30	1	manure	X070-X190-X002-XXXX
W4	30	30	1	manure	X070-X190-X002-XXXX
W5	76	76	2	manure	X070-X160-X045-XXXX
W6	76	76	1	manure	X070-X160-X045-XXXX
W7	65	65	2	manure	X070-X160-X056-XXXX
W8	72	72	2	manure	X070-X160-X057-XXXX
Totals for areas that were used for application	1,256	1,256	34		
Totals for areas that were not used for application	47	47	1		
Land application area totals	1,303	1,303	35		

B. CROPS AND HARVESTS

N1

Field name: N1

11/01/2020: Pasture - Grazed

Crop: Pasture - Grazed Acres planted: 16 Plant date: 11/01/2020

No harvests entered for this crop.

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.00	60.00	0.20	0.20	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

N10

Field name: N10

11/14/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 16 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	127.00 *ton*	Dry-weight		9.2	36,900.00	3,200.00	24,700.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.94	531.90	46.13	356.04	1,441.45

N11

Field name: N11

10/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 68 Plant date: 10/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	900.00 *ton*	Dry-weight		49.1	9,000.00	1,700.00	13,100.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.24	121.26	22.91	176.50	1,023.99

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,679.00 *ton*	Dry-weight		65.7	10,100.00	1,400.00	8,800.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.69	171.08	23.71	149.06	863.85

N2

Field name: N2

10/13/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 31 Plant date: 10/13/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	213.00 *ton*	Dry-weight		7.8	36,200.00	2,600.00	23,800.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	6.87	458.66	32.94	301.55	1,355.70

N3

Field name: N3

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 102 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	1,401.00 *ton*	Dry-weight		75.1	15,700.00	2,800.00	24,600.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.74	107.39	19.15	168.27	601.94

N4 (Picardo)

Field name: N4 (Picardo)

10/15/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 39 Plant date: 10/15/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	290.00 *ton*	Dry-weight		8.4	39,900.00	3,300.00	22,000.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.44	543.54	44.95	299.70	1,416.75

N5

Field name: N5

N5

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 62 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	445.00 *ton*	Dry-weight		7.6	43,400.00	3,900.00	25,200.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.18	575.65	51.73	334.25	1,326.39

N6

Field name: N6

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 37 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	268.00 *ton*	Dry-weight		7.6	43,400.00	3,900.00	25,200.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.24	580.93	52.20	337.31	1,338.55

N7

Field name: N7

N7

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 20 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	145.00 *ton*	Dry-weight		7.9	44,100.00	4,000.00	23,500.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.25	588.93	53.42	313.83	1,442.29

N8

Field name: N8

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 15 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	110.00 *ton*	Dry-weight		7.9	44,100.00	4,000.00	23,500.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.33	595.70	54.03	317.44	1,458.86

N9

Field name: N9

N9

11/14/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 23 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	183.00 *ton*	Dry-weight		9.2	36,900.00	3,200.00	24,700.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.96	533.17	46.24	356.89	1,444.90

W1

Field name: W1

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	851.00 *ton*	Dry-weight		55.4	13,300.00	2,600.00	19,400.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.95	165.51	32.35	241.42	1,095.08

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	1,424.00 *ton*	Dry-weight		64.5	11,900.00	2,600.00	9,900.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.34	197.24	43.09	164.09	828.72

W13

Field name: W13

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 82 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	983.00 *ton*	Dry-weight		58.2	10,000.00	2,200.00	17,100.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.99	100.22	22.05	171.37	831.81

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 82 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	1,995.00 *ton*	Dry-weight		64.5	12,400.00	2,400.00	8,800.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.33	214.19	41.46	152.01	811.87

W14

Field name: W14

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 87 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	1,079.00 *ton*	Dry-weight		58.2	10,000.00	2,200.00	17,100.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.40	103.68	22.81	177.30	860.57

W14

06/28/2023: Sudangrass, hay

Crop: Sudangrass, hay Acres planted: 87 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	406.00 *ton*	Dry-weight		8.4	15,500.00	3,600.00	30,700.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	4.67	132.51	30.78	262.46	1,205.46

W15

Field name: W15

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 48 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	614.00 *ton*	Dry-weight		62.6	18,800.00	3,100.00	25,200.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.79	179.88	29.66	241.12	832.43

06/19/2023: Sudangrass, hay

Crop: Sudangrass, hay Acres planted: 48 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	224.00 *ton*	Dry-weight		8.3	15,000.00	3,800.00	33,100.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	4.67	128.38	32.52	283.29	1,206.77

W16

Field name: W16

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	960.00 *ton*	Dry-weight		62.6	18,800.00	3,100.00	25,200.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.80	180.00	29.68	241.27	832.97

06/19/2023: Sudangrass, hay

Crop: Sudangrass, hay Acres planted: 75 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	351.00 *ton*	Dry-weight		8.6	15,000.00	4,000.00	30,200.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	256.00	35.20	264.00	0.00
Total actual harvest content	4.68	128.33	34.22	258.36	1,274.70

W2

Field name: W2

10/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 10/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	830.00 *ton*	Dry-weight		63.8	13,200.00	4,100.00	19,900.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.83	132.20	41.06	199.31	901.38

W2

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,428.00 *ton*	Dry-weight		67.8	11,100.00	3,200.00	9,900.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.80	170.13	49.05	151.74	873.65

W20

Field name: W20

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	525.00 *ton*	Dry-weight		71.4	15,800.00	2,600.00	21,900.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.46	121.66	20.02	168.63	700.70

W23

Field name: W23

W23

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough | Acres planted: 73 | Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	1,162.00 *ton*	Dry-weight		76.0	18,300.00	3,200.00	28,800.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.92	139.82	24.45	220.05	748.77

05/22/2023: Corn, silage

Crop: Corn, silage | Acres planted: 73 | Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	1,799.00 *ton*	Dry-weight		65.9	9,400.00	2,300.00	8,500.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.64	157.99	38.66	142.86	840.35

W3

Field name: W3

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay | Acres planted: 30 | Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	215.00 *ton*	Dry-weight		8.1	42,000.00	3,800.00	24,300.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.17	553.24	50.05	320.09	1,159.17

W4

Field name: W4

11/15/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 30 Plant date: 11/15/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	216.00 *ton*	Dry-weight		6.7	20,500.00	2,500.00	26,400.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.20	275.42	33.59	354.69	980.77

W5

Field name: W5

10/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 76 Plant date: 10/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	924.00 *ton*	Dry-weight		61.1	11,100.00	2,700.00	19,000.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.16	104.99	25.54	179.72	860.75

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 76 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,842.00 *ton*	Dry-weight		66.4	11,000.00	2,900.00	9,900.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.24	179.16	47.23	161.24	960.94

W6

Field name: W6

10/05/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 76 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	600.00 *ton*	Dry-weight		8.0	35,400.00	2,800.00	24,000.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.89	514.23	40.67	348.63	1,539.79

W7

Field name: W7

10/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 10/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	676.00 *ton*	Dry-weight		65.0	21,700.00	3,500.00	24,600.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.40	157.98	25.48	179.09	662.48

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,526.00 *ton*	Dry-weight		65.4	11,500.00	2,200.00	10,100.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.48	186.83	35.74	164.08	877.29

W8

Field name: W8

10/15/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 50 Plant date: 10/15/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	370.00 *ton*	Dry-weight		7.9	41,100.00	3,300.00	24,600.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.40	560.23	44.98	335.32	1,322.19

10/15/2021: Jose Grass

Crop: Jose Grass Acres planted: 22 Plant date: 10/15/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	105.20 *ton*	Dry-weight		65.2	25,300.00	2,700.00	29,000.00		12.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	270.00	18.00	18.00	0.00
Total actual harvest content	4.78	84.20	8.99	96.52	415.35

NUTRIENT BUDGET

A. LAND APPLICATIONS

N1 - 11/01/2020: Pasture - Grazed

Field name: N1

Crop: Pasture - Grazed

Plant date: 11/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	3,301,709.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	3,148,738.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	2,563,654.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	4,679,561.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	3,158,787.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

N1 - 11/01/2020: Pasture - Grazed

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	2,218,632.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.06	2,119,257.00 *gal*
Application event totals		0.77	0.00	0.00	221.06	

N10 - 11/14/2022: Alfalfa, hay

Field name: N10

Crop: Alfalfa, hay

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	292.47	62.91	295.87	0.00	208.00 *ton*
Application event totals		292.47	62.91	295.87	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	3,301,709.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	3,148,738.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

N10 - 11/14/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	2,563,654.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	4,679,561.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	3,158,787.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	2,218,632.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.06	2,119,257.00 *gal*
Application event totals		0.77	0.00	0.00	221.06	

N11 - 10/06/2022: Oats, silage-soft dough

Field name: N11

Crop: Oats, silage-soft dough

Plant date: 10/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

N11 - 10/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	160.85	23.21	91.23	0.00	935.00 *ton*
Application event totals		160.85	23.21	91.23	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	12,048,164.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	8,737,239.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

N11 - 06/30/2023: Corn, silage

Field name: N11

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	13,918,509.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	1.25	0.00	0.00	356.17	14,511,600.00 *gal*
Application event totals		81.25	0.00	0.00	356.17	

N11 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.98	0.00	0.00	278.85	11,361,202.00 *gal*
Application event totals		80.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	14,664,656.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.64	0.00	0.00	2,274.36	12,960,000.00 *gal*
Application event totals		0.64	0.00	0.00	2,274.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	11,951,105.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	10,171,831.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

N2 - 10/13/2018: Alfalfa, hay

Field name: N2

Crop: Alfalfa, hay

Plant date: 10/13/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

N2 - 10/13/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	6,397,061.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	6,100,681.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	4,967,079.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	9,066,650.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	6,120,151.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	4,298,600.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

N2 - 10/13/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	4,106,061.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

N3 - 10/11/2022: Oats, silage-soft dough

Field name: N3

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.03	6.60	131.81	716.79	1,176,000.00 *gal*
Canal	Surface water	1.03	0.00	0.00	295.71	18,072,247.00 *gal*
Application event totals		71.07	6.60	131.81	1,012.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.15	4.81	101.02	582.09	1,176,000.00 *gal*
Canal	Surface water	0.75	0.00	0.00	214.45	13,105,858.00 *gal*
Application event totals		63.90	4.81	101.02	796.54	

N4 (Picardo) - 10/15/2021: Alfalfa, hay

Field name: N4 (Picardo)

Crop: Alfalfa, hay

Plant date: 10/15/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

N4 (Picardo) - 10/15/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	8,047,916.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	7,675,050.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	6,248,906.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	11,406,430.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	7,699,545.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	5,407,916.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

N4 (Picardo) - 10/15/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.06	5,165,689.00 *gal*
Application event totals		0.77	0.00	0.00	221.06	

10/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	183.17	29.09	89.43	0.00	1,015.00 *ton*
Application event totals		183.17	29.09	89.43	0.00	

N5 - 11/15/2020: Alfalfa, hay

Field name: N5

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	12,794,122.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	12,201,361.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	9,934,158.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

N5 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	18,133,299.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	12,240,302.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	8,597,200.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	8,212,122.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	183.33	29.12	89.51	0.00	1,615.00 *ton*
Application event totals		183.33	29.12	89.51	0.00	

N6 - 11/15/2020: Alfalfa, hay

Field name: N6

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

N6 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	7,635,202.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	7,281,457.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	5,928,449.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	10,821,485.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	7,304,696.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	5,130,587.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

N6 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	4,900,782.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

10/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	182.61	29.00	89.16	0.00	960.00 *ton*
Application event totals		182.61	29.00	89.16	0.00	

N7 - 11/15/2020: Alfalfa, hay

Field name: N7

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	4,127,136.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	3,935,923.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	3,204,567.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

N7 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	5,849,451.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	3,948,484.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	2,773,290.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	2,649,072.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.75	29.34	90.20	0.00	525.00 *ton*
Application event totals		184.75	29.34	90.20	0.00	

N8 - 11/15/2020: Alfalfa, hay

Field name: N8

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

N8 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	3,095,352.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	2,951,942.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	2,403,425.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	4,387,089.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	2,961,363.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	2,079,968.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

N8 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	1,986,804.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

10/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	187.68	29.81	91.63	0.00	400.00 *ton*
Application event totals		187.68	29.81	91.63	0.00	

N9 - 11/14/2022: Alfalfa, hay

Field name: N9

Crop: Alfalfa, hay

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	292.47	62.91	295.87	0.00	299.00 *ton*
Application event totals		292.47	62.91	295.87	0.00	

04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	4,746,207.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	4,526,311.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

N9 - 11/14/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	3,685,252.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	6,726,869.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	4,540,757.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	3,189,284.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	3,046,432.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

W1 - 10/24/2022: Oats, silage-soft dough

Field name: W1

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W1 - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	10,807,912.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	7,837,817.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

W1 - 05/28/2023: Corn, silage

Field name: W1

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	224.11	51.13	123.56	2,147.37	1,130.00 *ton*
Application event totals		224.11	51.13	123.56	2,147.37	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	12,485,721.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	356.17	13,017,759.00 *gal*
Application event totals		1.25	0.00	0.00	356.17	

W1 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	278.85	10,191,667.00 *gal*
Application event totals		0.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	13,155,059.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.71	0.00	0.00	2,535.35	12,960,000.00 *gal*
Application event totals		0.71	0.00	0.00	2,535.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	10,720,844.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	9,124,731.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W13 - 10/08/2022: Oats, silage-soft dough

Field name: W13

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W13 - 10/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	130.53	18.83	74.03	0.00	915.00 *ton*
Application event totals		130.53	18.83	74.03	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	14,528,669.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	10,536,082.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

W13 - 05/28/2023: Corn, silage

Field name: W13

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	16,784,084.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	1.25	0.00	0.00	356.17	17,499,283.00 *gal*
Application event totals		81.25	0.00	0.00	356.17	

W13 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	278.85	13,700,273.00 *gal*
Application event totals		0.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	1.26	0.00	0.00	359.93	17,683,850.00 *gal*
Application event totals		81.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.29	0.00	0.00	6,960.95	14,400,000.00 *gal*
Application event totals		0.29	0.00	0.00	6,960.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	14,411,626.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	12,266,032.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W14 - 10/08/2022: Oats, silage-soft dough

Field name: W14

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W14 - 10/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	136.48	19.69	77.41	0.00	1,015.00 *ton*
Application event totals		136.48	19.69	77.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	15,414,563.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	11,178,526.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

W14 - 06/28/2023: Sudangrass, hay

Field name: W14

Crop: Sudangrass, hay

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	17,807,504.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Canal	Surface water	1.25	0.00	0.00	356.17	18,566,312.00 *gal*
Application event totals		71.25	0.00	0.00	356.17	

W14 - 06/28/2023: Sudangrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Canal	Surface water	0.98	0.00	0.00	278.85	14,535,656.00 *gal*
Application event totals		70.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	18,762,133.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.28	0.00	0.00	6,560.90	14,400,000.00 *gal*
Application event totals		0.28	0.00	0.00	6,560.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	15,290,384.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	13,013,961.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W15 - 10/10/2022: Oats, silage-soft dough

Field name: W15

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W15 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.67	9.02	180.06	979.18	756,000.00 *gal*
Canal	Surface water	1.03	0.00	0.00	295.71	8,504,587.00 *gal*
Application event totals		96.71	9.02	180.06	1,274.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.86	7.30	153.34	883.53	840,000.00 *gal*
Freshwater Pond	Surface water	0.47	0.00	0.00	0.00	5,388,000.00 *gal*
Application event totals		96.33	7.30	153.34	883.53	

W15 - 06/19/2023: Sudangrass, hay

Field name: W15

Crop: Sudangrass, hay

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	9,824,830.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.53	14.60	205.62	838.84	672,000.00 *gal*
Canal	Surface water	1.25	0.00	0.00	356.17	10,243,482.00 *gal*
Application event totals		90.77	14.60	205.62	1,195.01	

W15 - 06/19/2023: Sudangrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	278.85	8,019,672.00 *gal*
Application event totals		0.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.74	11.87	167.07	681.56	546,000.00 *gal*
Canal	Surface water	1.26	0.00	0.00	359.93	10,351,522.00 *gal*
Application event totals		74.00	11.87	167.07	1,041.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	0.75	0.00	0.00	0.00	8,620,800.00 *gal*
Application event totals		0.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	8,436,074.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	7,180,116.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W16 - 10/10/2022: Oats, silage-soft dough

Field name: W16

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W16 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.25	8.98	179.26	974.83	1,176,000.00 *gal*
Canal	Surface water	1.03	0.00	0.00	295.71	13,288,417.00 *gal*
Application event totals		96.28	8.98	179.26	1,270.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.16	7.48	157.02	904.73	1,344,000.00 *gal*
Freshwater Pond	Surface water	0.60	0.00	0.00	0.00	10,776,000.00 *gal*
Application event totals		98.76	7.48	157.02	904.73	

W16 - 06/19/2023: Sudangrass, hay

Field name: W16

Crop: Sudangrass, hay

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.30	13.61	96.83	758.93	1,176,000.00 *gal*
Canal	Surface water	1.20	0.00	0.00	341.62	15,351,297.00 *gal*
Application event totals		69.50	13.61	96.83	1,100.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	356.17	16,005,441.00 *gal*
Application event totals		1.25	0.00	0.00	356.17	

W16 - 06/19/2023: Sudangrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.95	14.02	197.40	805.29	1,008,000.00 *gal*
Canal	Surface water	0.98	0.00	0.00	278.85	12,530,738.00 *gal*
Application event totals		86.92	14.02	197.40	1,084.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	16,174,253.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	0.96	0.00	0.00	0.00	17,241,600.00 *gal*
Application event totals		0.96	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	13,181,365.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	11,218,932.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W2 - 10/06/2022: Oats, silage-soft dough

Field name: W2

Crop: Oats, silage-soft dough

Plant date: 10/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W2 - 10/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	10,630,733.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	7,709,328.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

W2 - 05/28/2023: Corn, silage

Field name: W2

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	223.81	51.06	123.40	2,144.52	1,110.00 *ton*
Application event totals		223.81	51.06	123.40	2,144.52	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	12,281,037.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	356.17	12,804,353.00 *gal*
Application event totals		1.25	0.00	0.00	356.17	

W2 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	278.85	10,024,590.00 *gal*
Application event totals		0.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	12,939,402.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.72	0.00	0.00	2,577.60	12,960,000.00 *gal*
Application event totals		0.72	0.00	0.00	2,577.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	10,545,092.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	8,975,145.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W20 - 10/10/2022: Oats, silage-soft dough

Field name: W20

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W20 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.96	6.78	135.43	736.48	462,000.00 *gal*
Canal	Surface water	1.03	0.00	0.00	295.71	6,909,977.00 *gal*
Application event totals		72.99	6.78	135.43	1,032.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.69	5.84	122.67	706.82	546,000.00 *gal*
Canal	Surface water	0.75	0.00	0.00	214.45	5,011,063.00 *gal*
Application event totals		77.44	5.84	122.67	921.27	

W23 - 10/11/2022: Oats, silage-soft dough

Field name: W23

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	184.29	26.59	104.52	0.00	1,150.00 *ton*
Application event totals		184.29	26.59	104.52	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	12,934,059.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

W23 - 10/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	0.62	0.00	0.00	0.00	10,776,000.00 *gal*
Application event totals		0.62	0.00	0.00	0.00	

W23 - 05/22/2023: Corn, silage

Field name: W23

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	14,941,929.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Canal	Surface water	1.25	0.00	0.00	356.17	15,578,630.00 *gal*
Application event totals		71.25	0.00	0.00	356.17	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN-32	Liquid commercial fertilizer	70.00	0.00	0.00	0.00	
Canal	Surface water	0.98	0.00	0.00	278.85	12,196,585.00 *gal*
Application event totals		70.98	0.00	0.00	278.85	

W23 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	15,742,939.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	0.99	0.00	0.00	0.00	17,241,600.00 *gal*
Application event totals		0.99	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	12,829,862.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	10,919,760.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W3 - 11/15/2020: Alfalfa, hay

Field name: W3

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	6,190,704.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

W3 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	5,903,884.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	4,806,851.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	8,774,177.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	5,922,727.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	4,159,936.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	3,973,607.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

W3 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	187.68	29.81	91.63	0.00	800.00 *ton*
Application event totals		187.68	29.81	91.63	0.00	

W4 - 11/15/2020: Alfalfa, hay

Field name: W4

Crop: Alfalfa, hay

Plant date: 11/15/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	6,190,704.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	5,903,884.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	4,806,851.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	8,774,177.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

W4 - 11/15/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	5,922,727.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	4,159,936.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	3,973,607.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	187.68	29.81	91.63	0.00	800.00 *ton*
Application event totals		187.68	29.81	91.63	0.00	

W5 - 10/06/2022: Oats, silage-soft dough

Field name: W5

Crop: Oats, silage-soft dough

Plant date: 10/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	137.76	19.88	78.13	0.00	895.00 *ton*
Application event totals		137.76	19.88	78.13	0.00	

W5 - 10/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	13,465,595.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	9,765,149.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

W5 - 05/24/2023: Corn, silage

Field name: W5

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	237.18	54.11	130.77	2,272.64	1,490.00 *ton*
Application event totals		237.18	54.11	130.77	2,272.64	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	15,555,981.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	356.17	16,218,847.00 *gal*
Application event totals		1.25	0.00	0.00	356.17	

W5 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	278.85	12,697,814.00 *gal*
Application event totals		0.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	16,389,910.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	4,401.15	13,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	4,401.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	13,357,117.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	11,368,517.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W6 - 10/05/2022: Alfalfa, hay

Field name: W6

Crop: Alfalfa, hay

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W6 - 10/05/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	292.47	62.91	295.87	0.00	988.00 *ton*
Application event totals		292.47	62.91	295.87	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	15,683,118.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	14,956,507.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	12,177,355.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	22,227,915.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	15,004,241.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

W6 - 10/05/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	10,538,503.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	10,066,472.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

W7 - 10/07/2022: Oats, silage-soft dough

Field name: W7

Crop: Oats, silage-soft dough

Plant date: 10/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	137.68	19.86	78.09	0.00	765.00 *ton*
Application event totals		137.68	19.86	78.09	0.00	

10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	295.71	11,516,628.00 *gal*
Application event totals		1.03	0.00	0.00	295.71	

03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	214.45	8,351,772.00 *gal*
Application event totals		0.75	0.00	0.00	214.45	

W7 - 05/25/2023: Corn, silage

Field name: W7

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	237.31	54.14	130.83	2,273.82	1,275.00 *ton*
Application event totals		237.31	54.14	130.83	2,273.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	341.62	13,304,457.00 *gal*
Application event totals		1.20	0.00	0.00	341.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.25	0.00	0.00	356.17	13,871,383.00 *gal*
Application event totals		1.25	0.00	0.00	356.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	278.85	10,859,973.00 *gal*
Application event totals		0.98	0.00	0.00	278.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	359.93	14,017,686.00 *gal*
Application event totals		1.26	0.00	0.00	359.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.67	0.00	0.00	2,379.33	12,960,000.00 *gal*
Application event totals		0.67	0.00	0.00	2,379.33	

W7 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	293.33	11,423,850.00 *gal*
Application event totals		1.03	0.00	0.00	293.33	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.87	0.00	0.00	249.66	9,723,074.00 *gal*
Application event totals		0.87	0.00	0.00	249.66	

W8 - 10/15/2021: Alfalfa, hay

Field name: W8

Crop: Alfalfa, hay

Plant date: 10/15/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	10,317,840.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	9,839,807.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	8,011,418.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

W8 - 10/15/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	14,623,628.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	9,871,211.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	6,933,226.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	6,622,679.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	263.22	41.81	128.51	0.00	1,870.00 *ton*
Application event totals		263.22	41.81	128.51	0.00	

W8 - 10/15/2021: Jose Grass

Field name: W8

Crop: Jose Grass

Plant date: 10/15/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

W8 - 10/15/2021: Jose Grass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.21	0.00	0.00	344.41	4,539,850.00 *gal*
Application event totals		1.21	0.00	0.00	344.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	328.45	4,329,515.00 *gal*
Application event totals		1.15	0.00	0.00	328.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.94	0.00	0.00	267.42	3,525,024.00 *gal*
Application event totals		0.94	0.00	0.00	267.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	488.14	6,434,397.00 *gal*
Application event totals		1.71	0.00	0.00	488.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	329.50	4,343,333.00 *gal*
Application event totals		1.15	0.00	0.00	329.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.81	0.00	0.00	231.43	3,050,619.00 *gal*
Application event totals		0.81	0.00	0.00	231.43	

W8 - 10/15/2021: Jose Grass

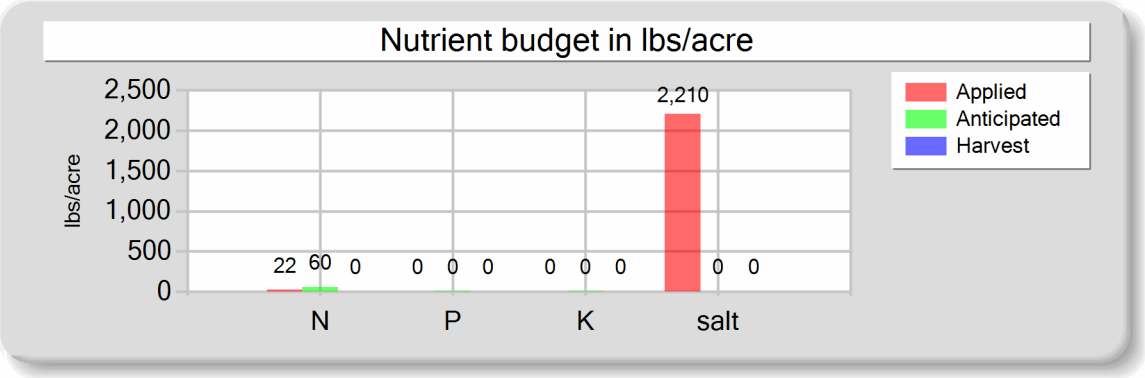
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.77	0.00	0.00	221.07	2,913,979.00 *gal*
Application event totals		0.77	0.00	0.00	221.07	

B. NUTRIENT BUDGET

N1 - 11/01/2020: Pasture - Grazed

Field name: N1 Crop: Pasture - Grazed Plant date: 11/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	21.74	0.00	0.00	2,210.42
Anticipated crop nutrient removal	60.00	0.20	0.20	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	21.74	0.00	0.00	2,210.42
Applied to removed ratio	0.00	0.00	0.00	0.00

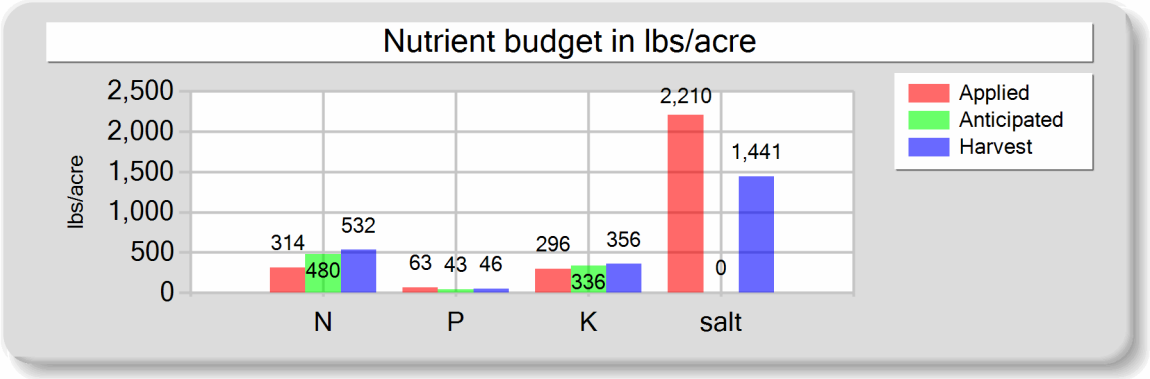
Fresh water applied	
21,190,338.00	*gallons*
780.37	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N10 - 11/14/2022: Alfalfa, hay

Field name: N10 Crop: Alfalfa, hay Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	292.47	62.91	295.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	314.21	62.91	295.87	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	531.90	46.13	356.04	1,441.45
Nutrient balance	-217.69	16.79	-60.17	768.97
Applied to removed ratio	0.59	1.36	0.83	1.53

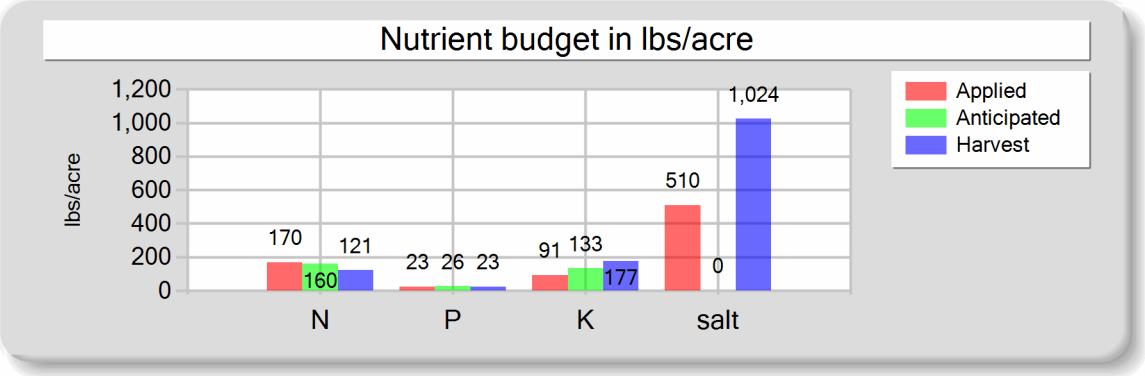
Fresh water applied	
21,190,338.00	*gallons*
780.37	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N11 - 10/06/2022: Oats, silage-soft dough

Field name: N11 Crop: Oats, silage-soft dough Plant date: 10/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	160.85	23.21	91.23	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.64	23.21	91.23	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.26	22.91	176.50	1,023.99
Nutrient balance	48.37	0.30	-85.27	-513.83
Applied to removed ratio	1.40	1.01	0.52	0.50

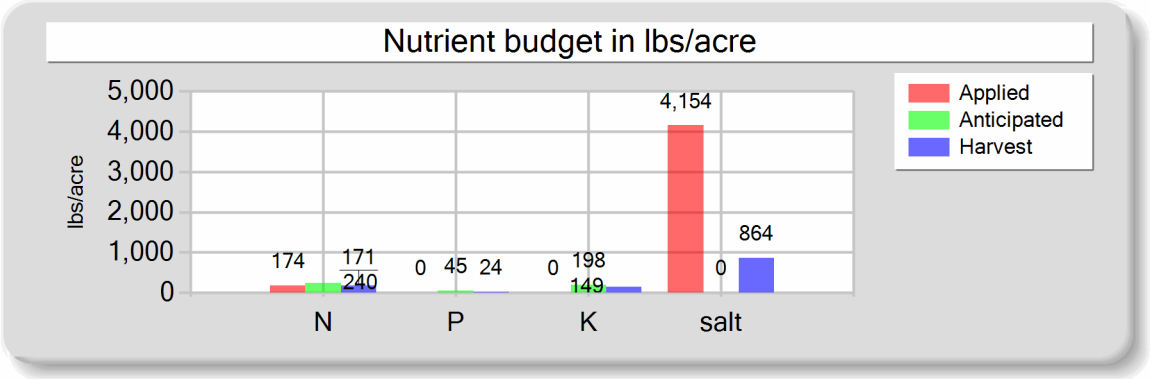
Fresh water applied	
20,785,403.00	*gallons*
765.46	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N11 - 06/30/2023: Corn, silage

Field name: N11 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	160.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.21	0.00	0.00	4,153.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.21	0.00	0.00	4,153.92
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	171.08	23.71	149.06	863.85
Nutrient balance	3.14	-23.71	-149.06	3,290.07
Applied to removed ratio	1.02	0.00	0.00	4.81

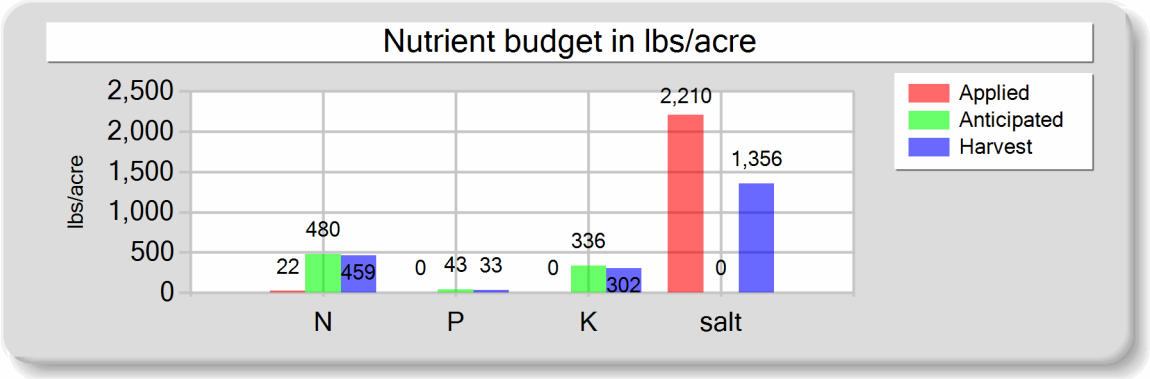
Fresh water applied	
89,538,903.00	*gallons*
3,297.41	*acre-inches*
48.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N2 - 10/13/2018: Alfalfa, hay

Field name: N2 Crop: Alfalfa, hay Plant date: 10/13/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	21.74	0.00	0.00	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	458.66	32.94	301.55	1,355.70
Nutrient balance	-436.92	-32.94	-301.55	854.72
Applied to removed ratio	0.05	0.00	0.00	1.63

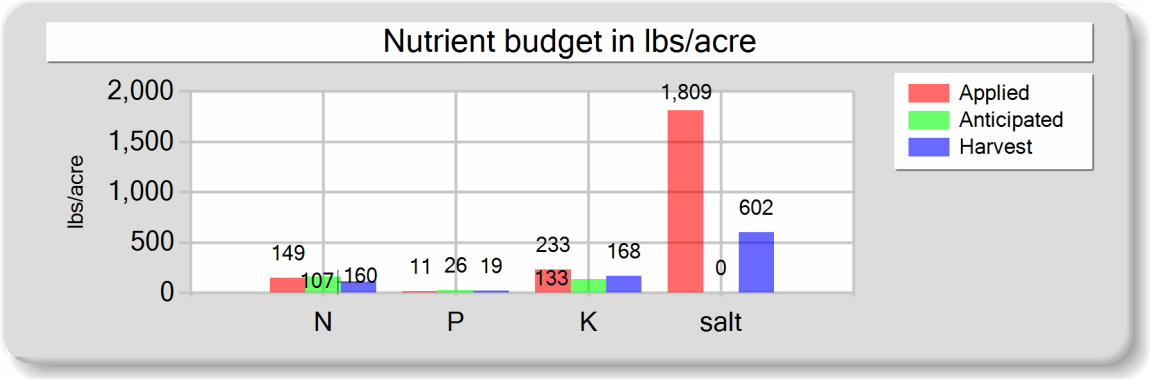
Fresh water applied	
41,056,283.00	*gallons*
1,511.96	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N3 - 10/11/2022: Oats, silage-soft dough

Field name: N3 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	133.19	11.41	232.84	1,298.87
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	148.97	11.41	232.84	1,809.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	107.39	19.15	168.27	601.94
Nutrient balance	41.58	-7.74	64.57	1,207.10
Applied to removed ratio	1.39	0.60	1.38	3.01

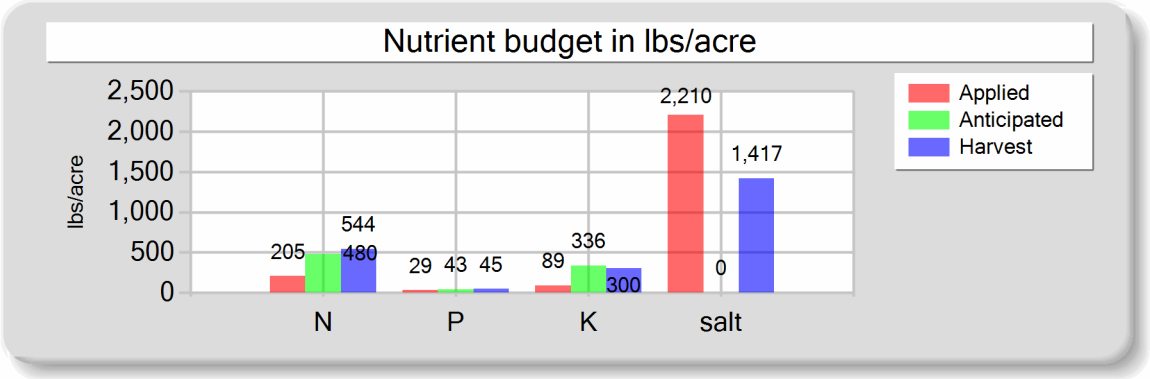
Fresh water applied	
31,178,105.00	*gallons*
1,148.18	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
2,352,000.00	*gallons*
86.62	*acre-inches*
0.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

N4 (Picardo) - 10/15/2021: Alfalfa, hay

Field name: N4 (Picardo) Crop: Alfalfa, hay Plant date: 10/15/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	183.17	29.09	89.43	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	204.90	29.09	89.43	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	543.54	44.95	299.70	1,416.75
Nutrient balance	-338.64	-15.86	-210.27	793.67
Applied to removed ratio	0.38	0.65	0.30	1.56

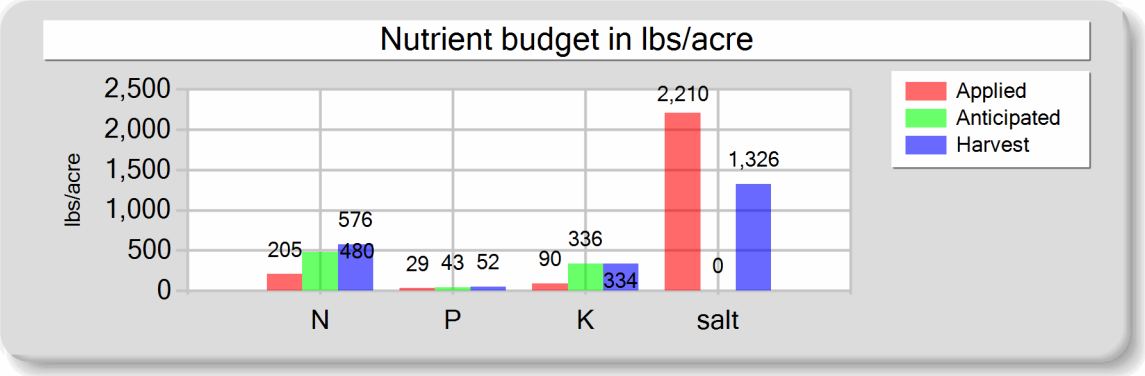
Fresh water applied	
51,651,452.00	*gallons*
1,902.15	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N5 - 11/15/2020: Alfalfa, hay

Field name: N5 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	183.33	29.12	89.51	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	205.07	29.12	89.51	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	575.65	51.73	334.25	1,326.39
Nutrient balance	-370.59	-22.61	-244.74	884.03
Applied to removed ratio	0.36	0.56	0.27	1.67

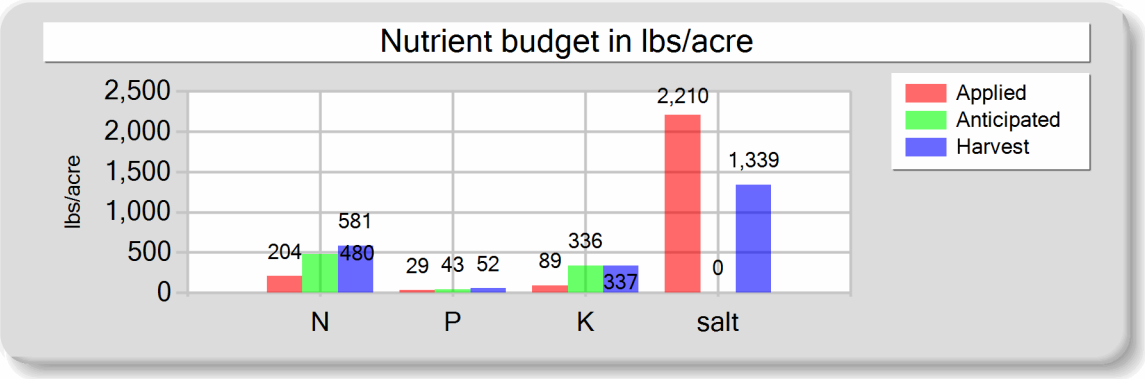
Fresh water applied	
82,112,564.00	*gallons*
3,023.93	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N6 - 11/15/2020: Alfalfa, hay

Field name: N6 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	182.61	29.00	89.16	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	204.34	29.00	89.16	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	580.93	52.20	337.31	1,338.55
Nutrient balance	-376.59	-23.20	-248.16	871.87
Applied to removed ratio	0.35	0.56	0.26	1.65

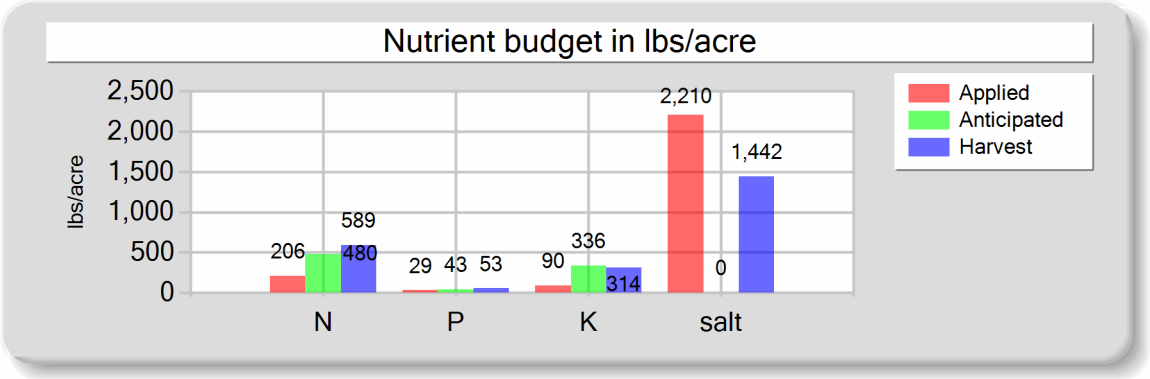
Fresh water applied	
49,002,658.00	*gallons*
1,804.60	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N7 - 11/15/2020: Alfalfa, hay

Field name: N7 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.75	29.34	90.20	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	206.48	29.34	90.20	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	588.93	53.42	313.83	1,442.29
Nutrient balance	-382.45	-24.08	-223.63	768.13
Applied to removed ratio	0.35	0.55	0.29	1.53

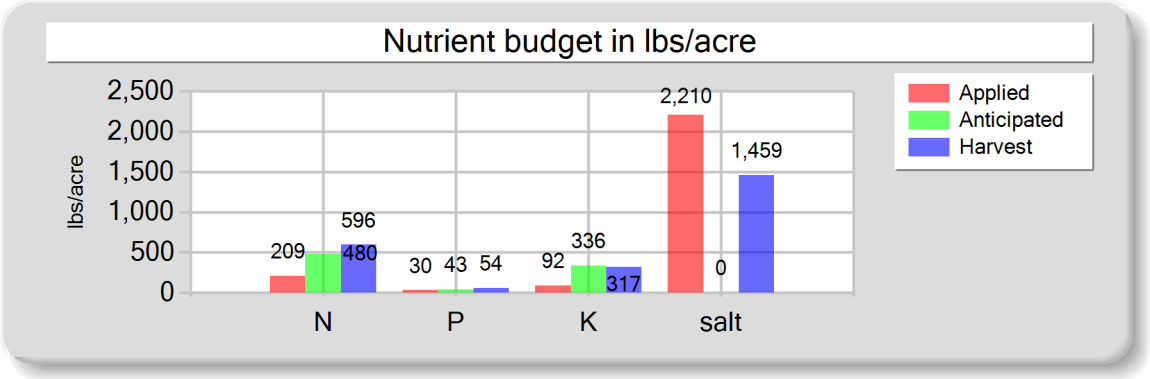
Fresh water applied	
26,487,923.00	*gallons*
975.46	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N8 - 11/15/2020: Alfalfa, hay

Field name: N8 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.68	29.81	91.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	209.42	29.81	91.63	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	595.70	54.03	317.44	1,458.86
Nutrient balance	-386.29	-24.22	-225.81	751.55
Applied to removed ratio	0.35	0.55	0.29	1.52

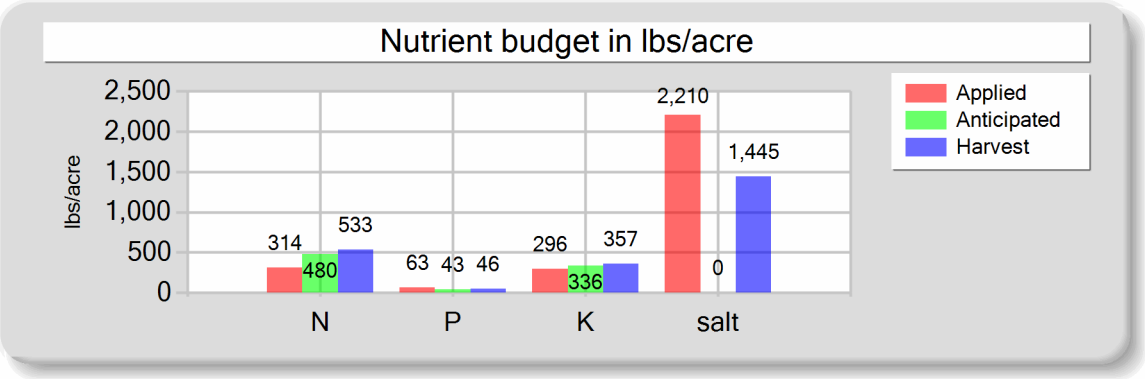
Fresh water applied	
19,865,943.00	*gallons*
731.60	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N9 - 11/14/2022: Alfalfa, hay

Field name: N9 Crop: Alfalfa, hay Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	292.47	62.91	295.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	314.21	62.91	295.87	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	533.17	46.24	356.89	1,444.90
Nutrient balance	-218.96	16.68	-61.02	765.51
Applied to removed ratio	0.59	1.36	0.83	1.53

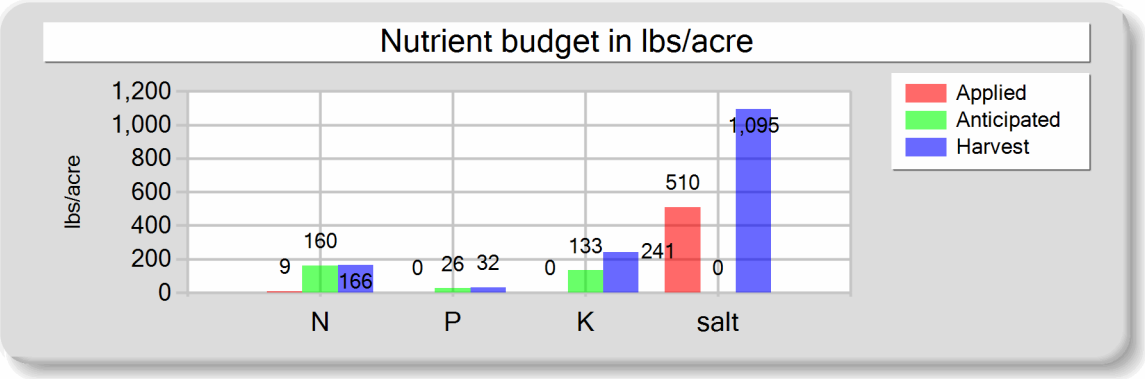
Fresh water applied	
30,461,112.00	*gallons*
1,121.78	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W1 - 10/24/2022: Oats, silage-soft dough

Field name: W1 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	8.79	0.00	0.00	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	165.51	32.35	241.42	1,095.08
Nutrient balance	-156.72	-32.35	-241.42	-584.92
Applied to removed ratio	0.05	0.00	0.00	0.47

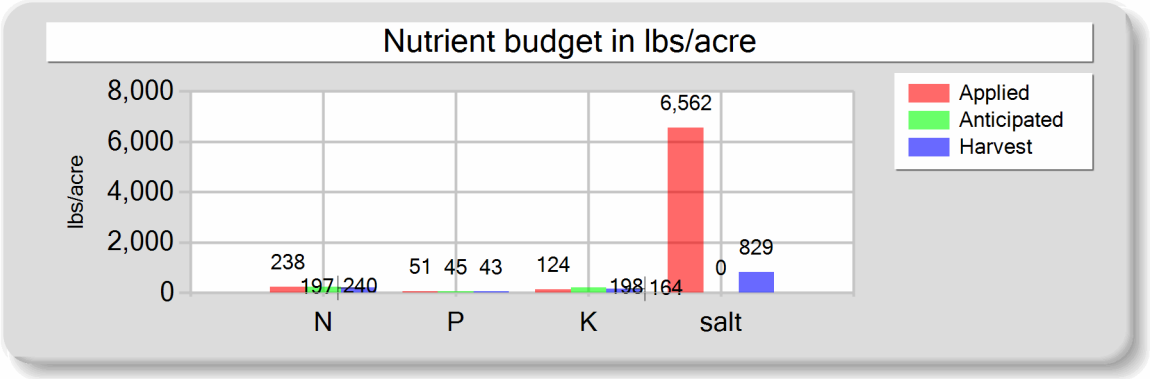
Fresh water applied	
18,645,729.00	*gallons*
686.66	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W1 - 05/28/2023: Corn, silage

Field name: W1 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	224.11	51.13	123.56	2,147.37
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.29	0.00	0.00	4,414.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.40	51.13	123.56	6,562.28
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	197.24	43.09	164.09	828.72
Nutrient balance	41.16	8.03	-40.53	5,733.56
Applied to removed ratio	1.21	1.19	0.75	7.92

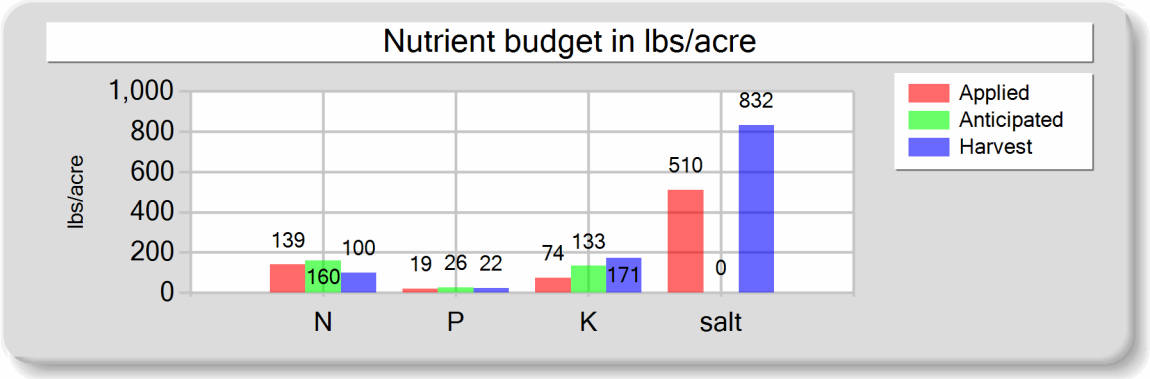
Fresh water applied	
81,655,781.00	*gallons*
3,007.10	*acre-inches*
49.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W13 - 10/08/2022: Oats, silage-soft dough

Field name: W13 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	130.53	18.83	74.03	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	139.32	18.83	74.03	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	100.22	22.05	171.37	831.81
Nutrient balance	39.10	-3.21	-97.34	-321.65
Applied to removed ratio	1.39	0.85	0.43	0.61

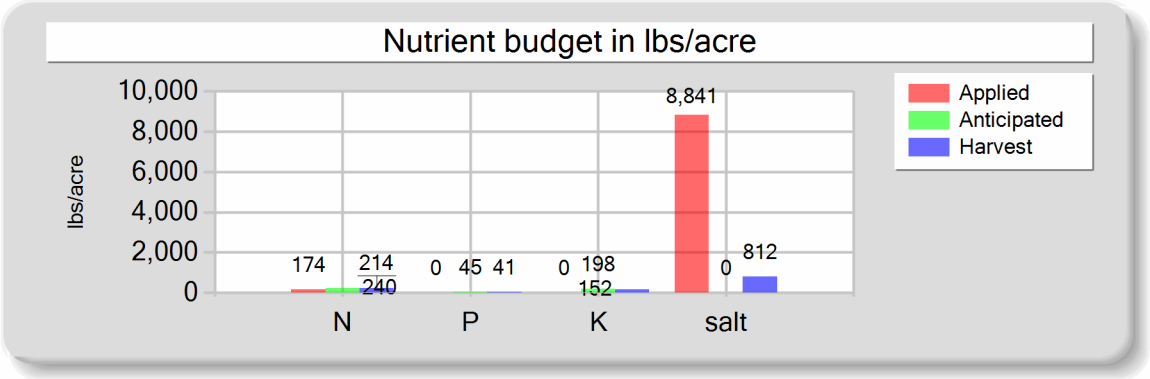
Fresh water applied	
25,064,751.00	*gallons*
923.05	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W13 - 05/28/2023: Corn, silage

Field name: W13 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	160.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.87	0.00	0.00	8,840.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.87	0.00	0.00	8,840.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	214.19	41.46	152.01	811.87
Nutrient balance	-40.32	-41.46	-152.01	8,028.65
Applied to removed ratio	0.81	0.00	0.00	10.89

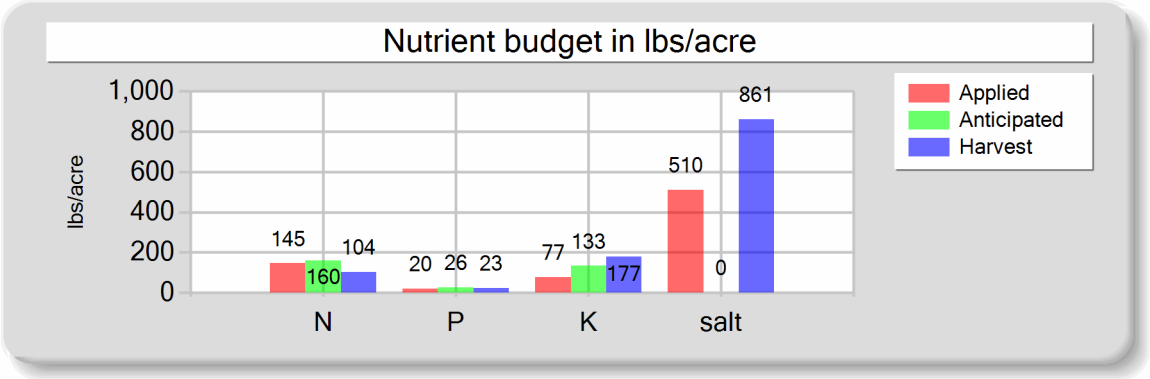
Fresh water applied	
106,745,148.00	*gallons*
3,931.06	*acre-inches*
47.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W14 - 10/08/2022: Oats, silage-soft dough

Field name: W14 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	136.48	19.69	77.41	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	145.26	19.69	77.41	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	103.68	22.81	177.30	860.57
Nutrient balance	41.58	-3.12	-99.89	-350.41
Applied to removed ratio	1.40	0.86	0.44	0.59

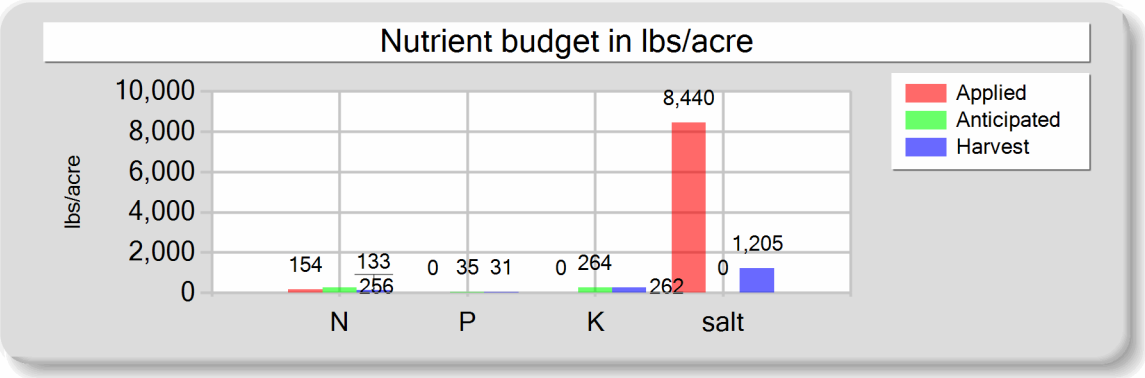
Fresh water applied	
26,593,089.00	*gallons*
979.33	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W14 - 06/28/2023: Sudangrass, hay

Field name: W14 Crop: Sudangrass, hay Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	140.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.85	0.00	0.00	8,440.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.85	0.00	0.00	8,440.46
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	132.51	30.78	262.46	1,205.46
Nutrient balance	21.34	-30.78	-262.46	7,235.00
Applied to removed ratio	1.16	0.00	0.00	7.00

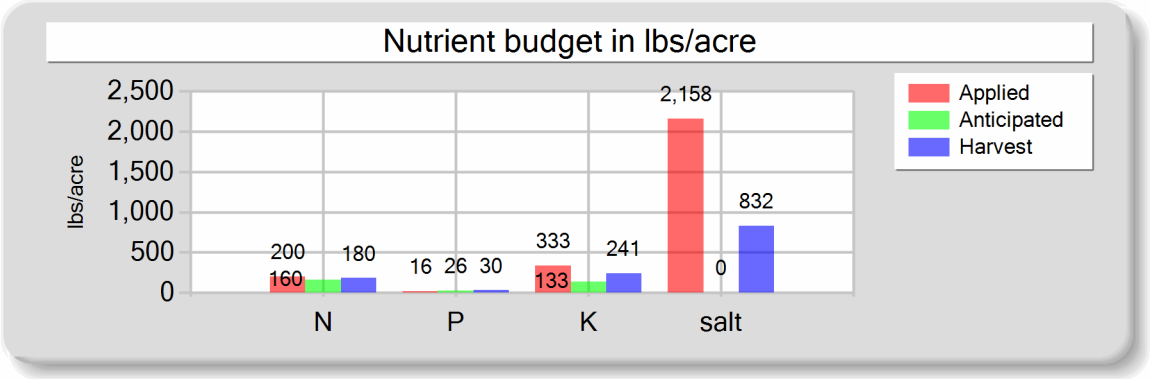
Fresh water applied	
112,375,950.00	*gallons*
4,138.42	*acre-inches*
47.57	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W15 - 10/10/2022: Oats, silage-soft dough

Field name: W15 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.53	16.32	333.40	1,862.71
Fresh water	1.50	0.00	0.00	295.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.03	16.32	333.40	2,158.42
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	179.88	29.66	241.12	832.43
Nutrient balance	20.15	-13.34	92.29	1,325.99
Applied to removed ratio	1.11	0.55	1.38	2.59

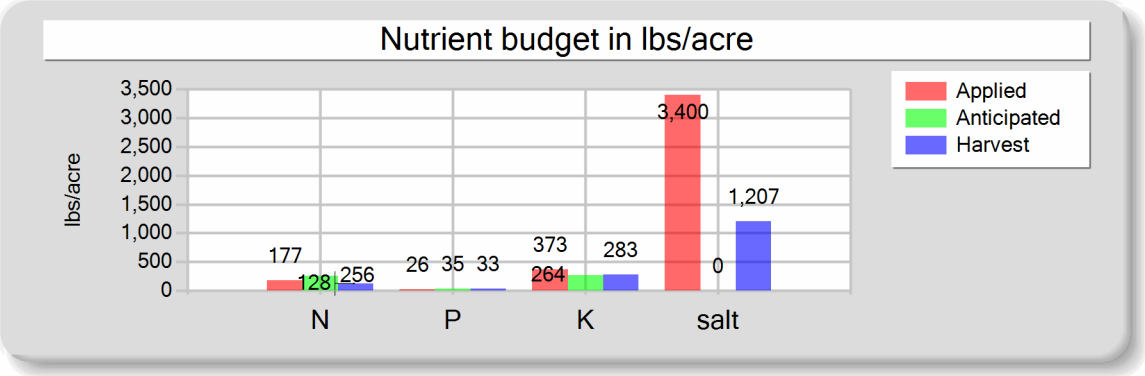
Fresh water applied	
13,892,587.00	*gallons*
511.62	*acre-inches*
10.66	*inches/acre*

Process wastewater applied	
1,596,000.00	*gallons*
58.78	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

W15 - 06/19/2023: Sudangrass, hay

Field name: W15 Crop: Sudangrass, hay Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	162.27	26.47	372.69	1,520.40
Fresh water	7.33	0.00	0.00	1,879.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.60	26.47	372.69	3,399.96
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	128.38	32.52	283.29	1,206.77
Nutrient balance	48.22	-6.05	89.40	2,193.19
Applied to removed ratio	1.38	0.81	1.32	2.82

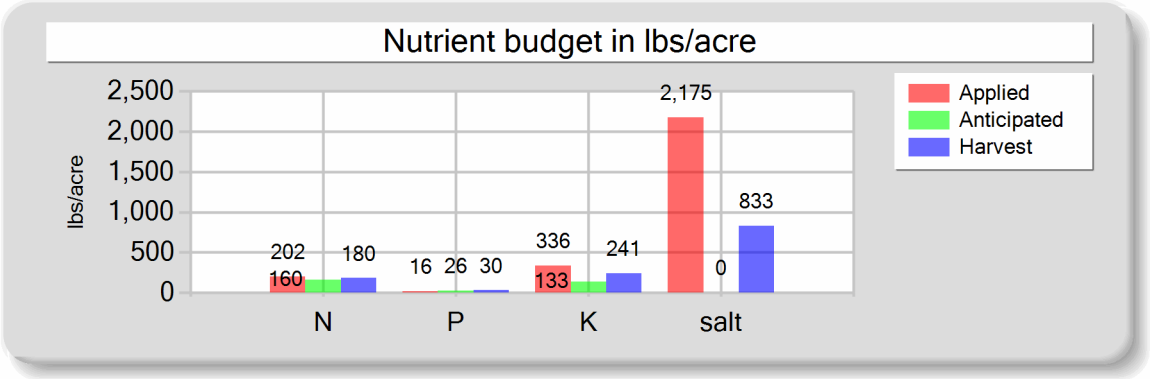
Fresh water applied	
62,676,496.00	*gallons*
2,308.16	*acre-inches*
48.09	*inches/acre*

Process wastewater applied	
1,218,000.00	*gallons*
44.85	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

W16 - 10/10/2022: Oats, silage-soft dough

Field name: W16 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.41	16.45	336.28	1,879.56
Fresh water	1.63	0.00	0.00	295.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.04	16.45	336.28	2,175.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	180.00	29.68	241.27	832.97
Nutrient balance	22.04	-13.23	95.01	1,342.30
Applied to removed ratio	1.12	0.55	1.39	2.61

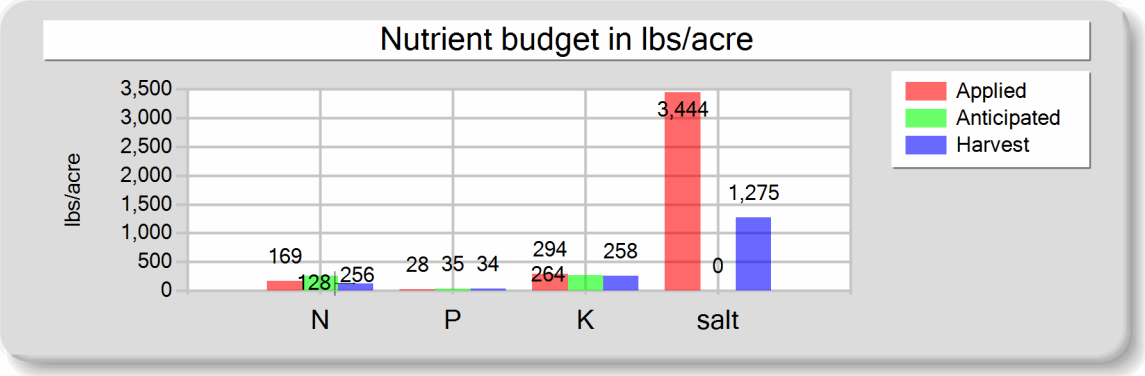
Fresh water applied	
24,064,417.00	*gallons*
886.21	*acre-inches*
11.82	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

W16 - 06/19/2023: Sudangrass, hay

Field name: W16 Crop: Sudangrass, hay Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.25	27.63	294.22	1,564.21
Fresh water	7.54	0.00	0.00	1,879.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.79	27.63	294.22	3,443.78
Anticipated crop nutrient removal	256.00	35.20	264.00	0.00
Actual crop nutrient removal	128.33	34.22	258.36	1,274.70
Nutrient balance	40.46	-6.59	35.86	2,169.07
Applied to removed ratio	1.32	0.81	1.14	2.70

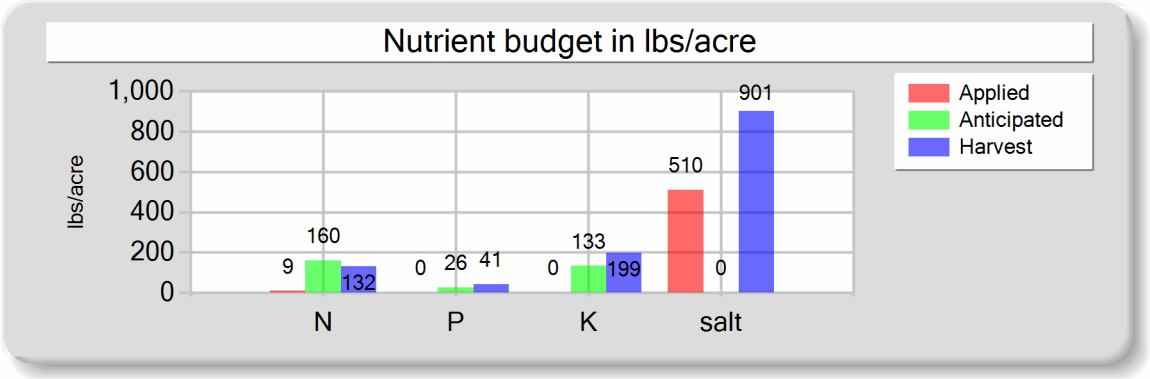
Fresh water applied	
101,703,626.00	*gallons*
3,745.40	*acre-inches*
49.94	*inches/acre*

Process wastewater applied	
2,184,000.00	*gallons*
80.43	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

W2 - 10/06/2022: Oats, silage-soft dough

Field name: W2 Crop: Oats, silage-soft dough Plant date: 10/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	8.79	0.00	0.00	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.20	41.06	199.31	901.38
Nutrient balance	-123.42	-41.06	-199.31	-391.22
Applied to removed ratio	0.07	0.00	0.00	0.57

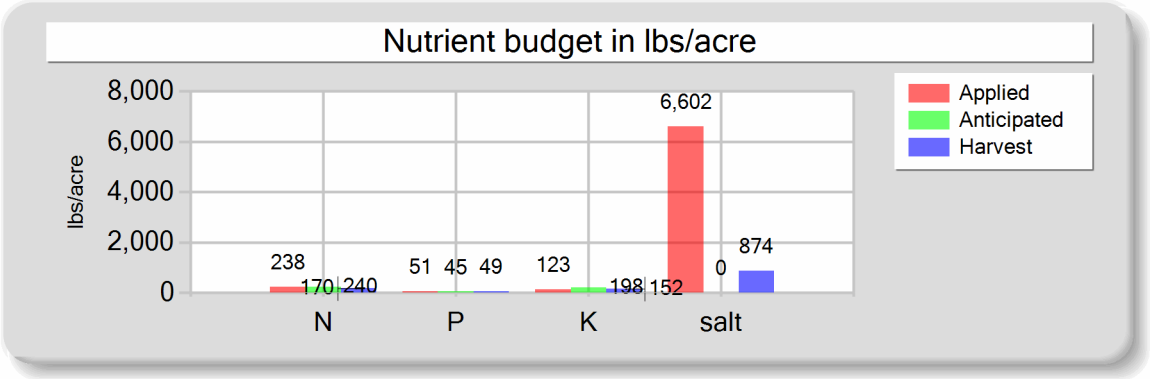
Fresh water applied	
18,340,061.00	*gallons*
675.40	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W2 - 05/28/2023: Corn, silage

Field name: W2 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	223.81	51.06	123.40	2,144.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.30	0.00	0.00	4,457.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.11	51.06	123.40	6,601.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	170.13	49.05	151.74	873.65
Nutrient balance	67.98	2.01	-28.34	5,728.03
Applied to removed ratio	1.40	1.04	0.81	7.56

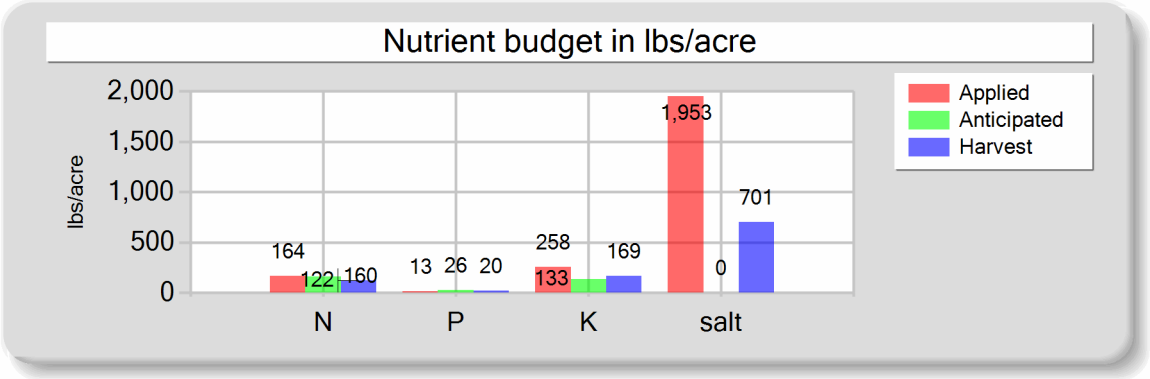
Fresh water applied	
80,529,619.00	*gallons*
2,965.63	*acre-inches*
49.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W20 - 10/10/2022: Oats, silage-soft dough

Field name: W20 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	148.64	12.62	258.10	1,443.30
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	164.43	12.62	258.10	1,953.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.66	20.02	168.63	700.70
Nutrient balance	42.77	-7.40	89.47	1,252.76
Applied to removed ratio	1.35	0.63	1.53	2.79

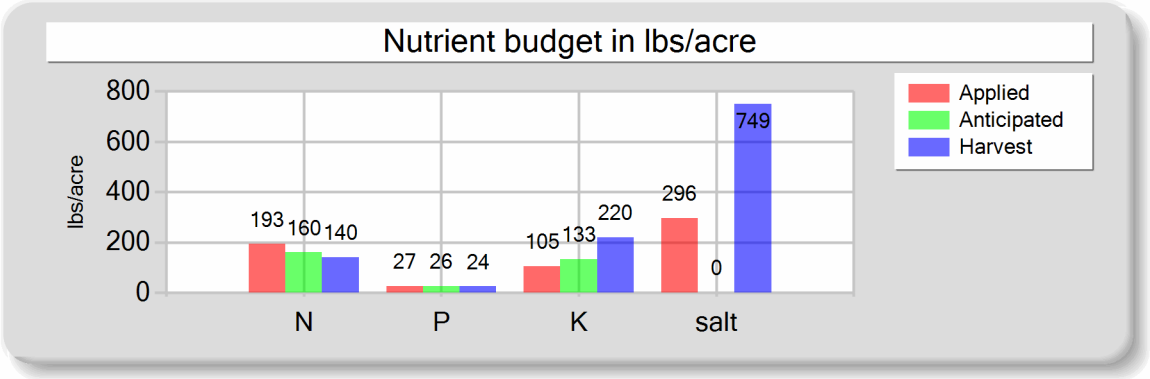
Fresh water applied	
11,921,040.00	*gallons*
439.01	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

W23 - 10/11/2022: Oats, silage-soft dough

Field name: W23 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	184.29	26.59	104.52	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.65	0.00	0.00	295.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.94	26.59	104.52	295.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.82	24.45	220.05	748.77
Nutrient balance	53.12	2.14	-115.53	-453.06
Applied to removed ratio	1.38	1.09	0.47	0.39

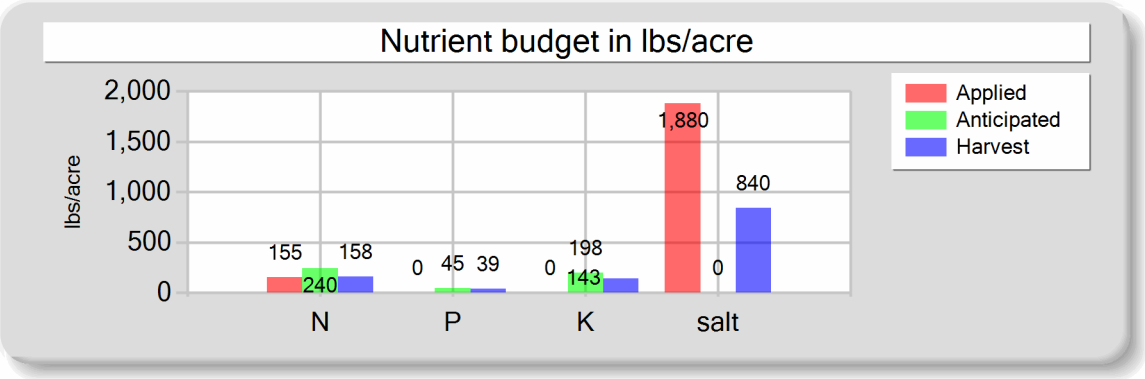
Fresh water applied	
23,710,059.00	*gallons*
873.16	*acre-inches*
11.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W23 - 05/22/2023: Corn, silage

Field name: W23 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	140.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.56	0.00	0.00	1,879.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.56	0.00	0.00	1,879.56
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	157.99	38.66	142.86	840.35
Nutrient balance	-3.42	-38.66	-142.86	1,039.21
Applied to removed ratio	0.98	0.00	0.00	2.24

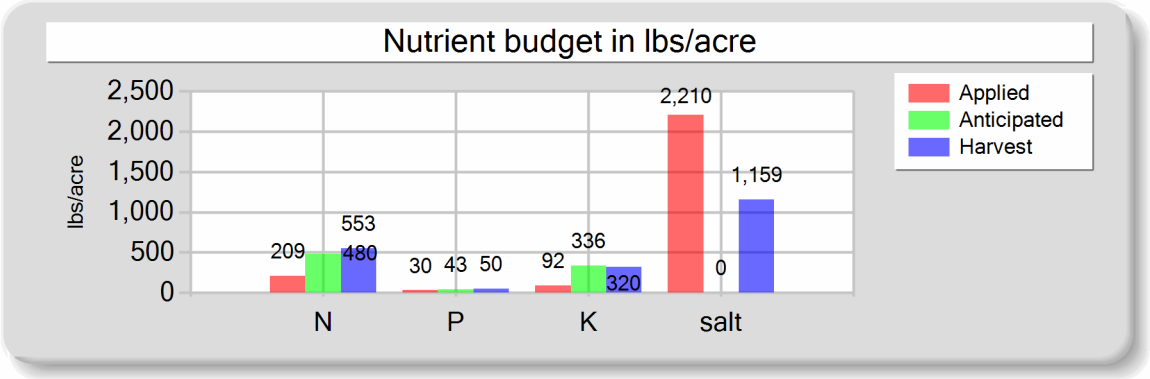
Fresh water applied	
99,451,305.00	*gallons*
3,662.45	*acre-inches*
50.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W3 - 11/15/2020: Alfalfa, hay

Field name: W3 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.68	29.81	91.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	209.42	29.81	91.63	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	553.24	50.05	320.09	1,159.17
Nutrient balance	-343.82	-20.25	-228.46	1,051.25
Applied to removed ratio	0.38	0.60	0.29	1.91

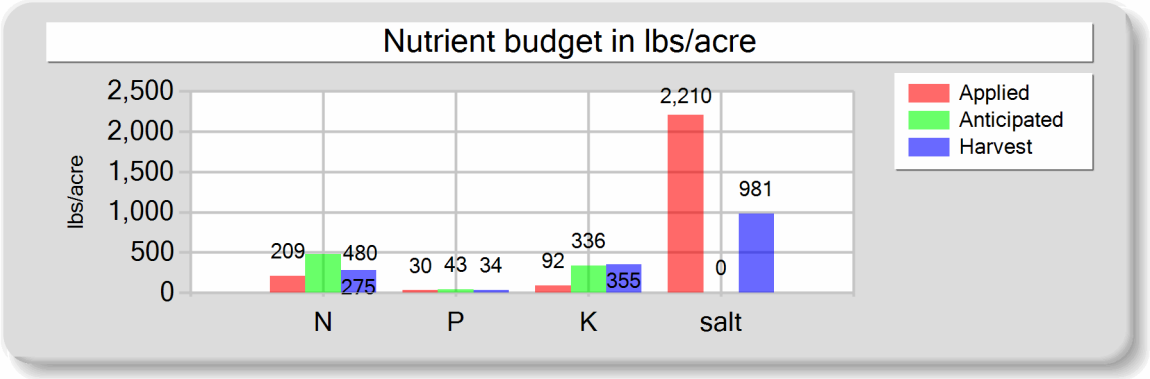
Fresh water applied	
39,731,886.00	*gallons*
1,463.19	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W4 - 11/15/2020: Alfalfa, hay

Field name: W4 Crop: Alfalfa, hay Plant date: 11/15/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.68	29.81	91.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	209.42	29.81	91.63	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	275.42	33.59	354.69	980.77
Nutrient balance	-66.01	-3.78	-263.06	1,229.65
Applied to removed ratio	0.76	0.89	0.26	2.25

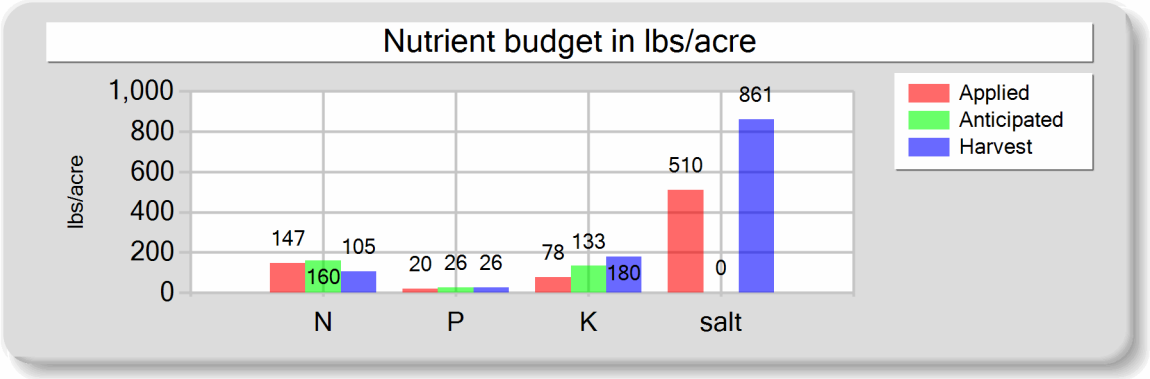
Fresh water applied	
39,731,886.00	*gallons*
1,463.19	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W5 - 10/06/2022: Oats, silage-soft dough

Field name: W5 Crop: Oats, silage-soft dough Plant date: 10/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.76	19.88	78.13	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.55	19.88	78.13	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	104.99	25.54	179.72	860.75
Nutrient balance	41.55	-5.66	-101.58	-350.60
Applied to removed ratio	1.40	0.78	0.43	0.59

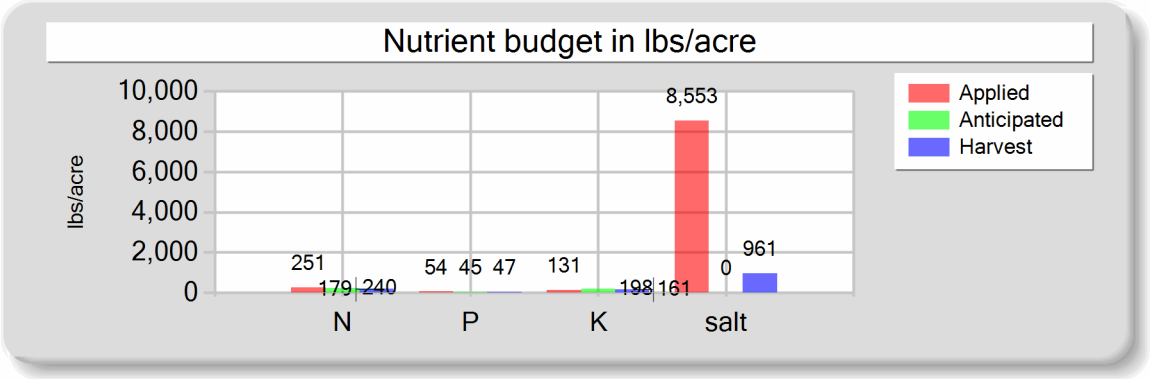
Fresh water applied	
23,230,744.00	*gallons*
855.51	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W5 - 05/24/2023: Corn, silage

Field name: W5 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	237.18	54.11	130.77	2,272.64
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.58	0.00	0.00	6,280.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.76	54.11	130.77	8,553.36
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	179.16	47.23	161.24	960.94
Nutrient balance	71.60	6.88	-30.48	7,592.41
Applied to removed ratio	1.40	1.15	0.81	8.90

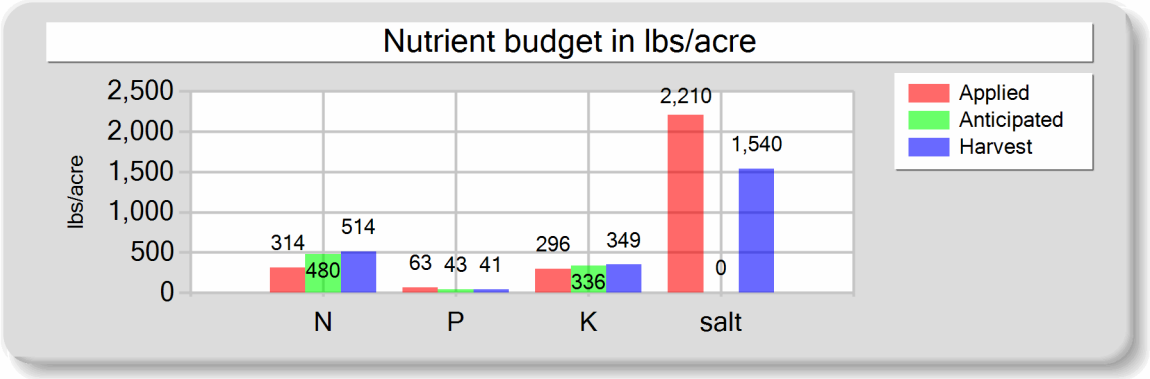
Fresh water applied	
99,268,186.00	*gallons*
3,655.71	*acre-inches*
48.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W6 - 10/05/2022: Alfalfa, hay

Field name: W6 Crop: Alfalfa, hay Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	292.47	62.91	295.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	314.21	62.91	295.87	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	514.23	40.67	348.63	1,539.79
Nutrient balance	-200.03	22.24	-52.76	670.63
Applied to removed ratio	0.61	1.55	0.85	1.44

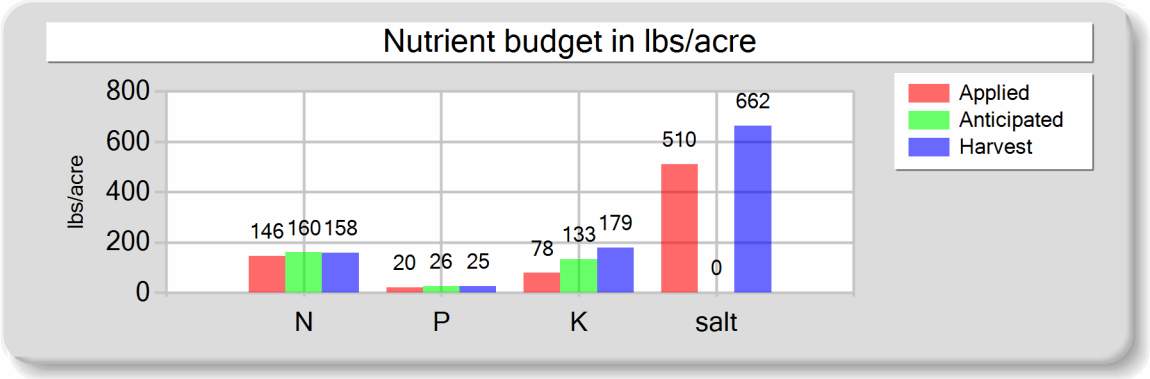
Fresh water applied	
100,654,111.00	*gallons*
3,706.75	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W7 - 10/07/2022: Oats, silage-soft dough

Field name: W7 Crop: Oats, silage-soft dough Plant date: 10/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.68	19.86	78.09	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.79	0.00	0.00	510.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.46	19.86	78.09	510.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.98	25.48	179.09	662.48
Nutrient balance	-11.51	-5.62	-101.00	-152.32
Applied to removed ratio	0.93	0.78	0.44	0.77

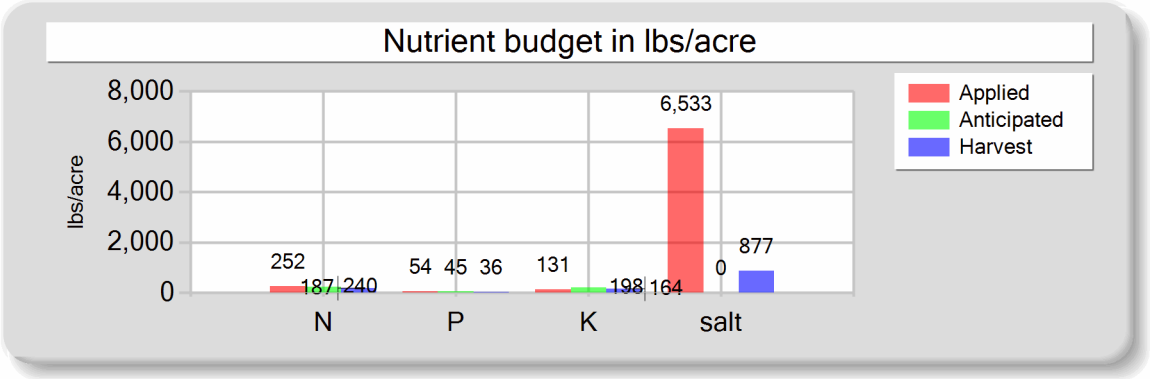
Fresh water applied	
19,868,400.00	*gallons*
731.69	*acre-inches*
11.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W7 - 05/25/2023: Corn, silage

Field name: W7 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	237.31	54.14	130.83	2,273.82
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.24	0.00	0.00	4,258.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.55	54.14	130.83	6,532.70
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	186.83	35.74	164.08	877.29
Nutrient balance	64.72	18.40	-33.25	5,655.42
Applied to removed ratio	1.35	1.51	0.80	7.45

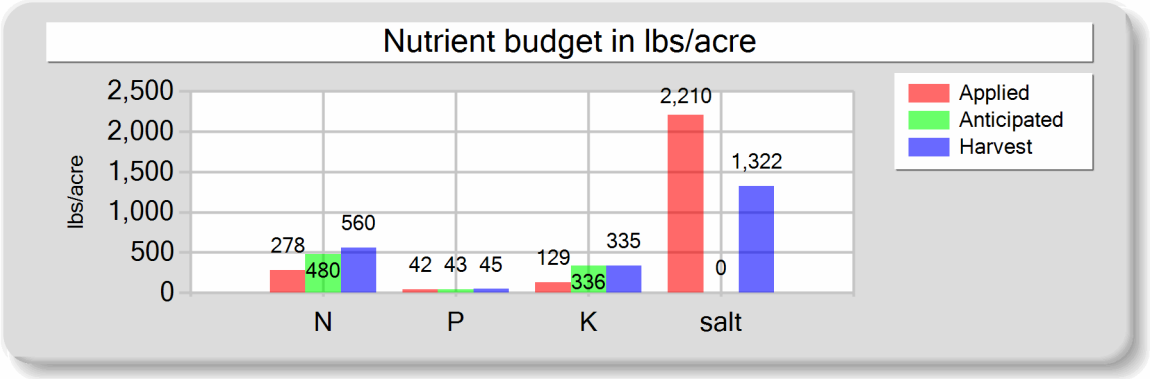
Fresh water applied	
86,160,423.00	*gallons*
3,173.00	*acre-inches*
48.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W8 - 10/15/2021: Alfalfa, hay

Field name: W8 Crop: Alfalfa, hay Plant date: 10/15/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	263.22	41.81	128.51	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.96	41.81	128.51	2,210.42
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	560.23	44.98	335.32	1,322.19
Nutrient balance	-282.27	-3.18	-206.80	888.23
Applied to removed ratio	0.50	0.93	0.38	1.67

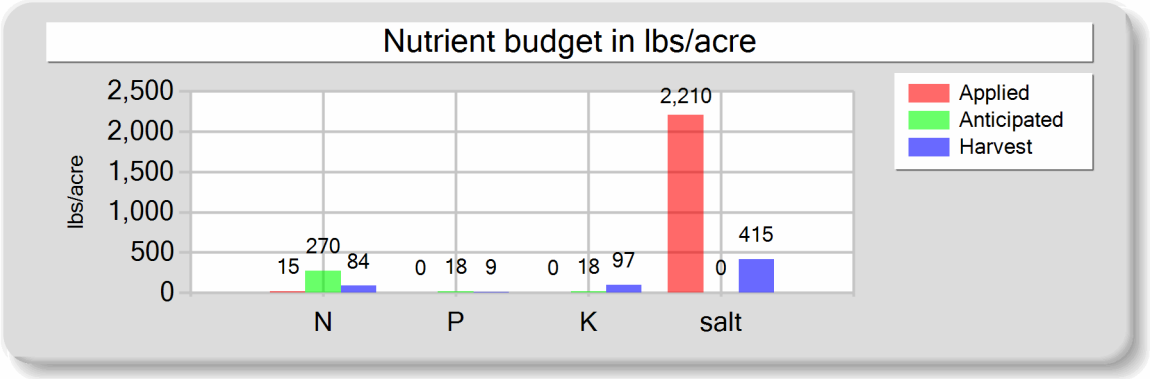
Fresh water applied	
66,219,809.00	*gallons*
2,438.65	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

W8 - 10/15/2021: Jose Grass

Field name: W8 Crop: Jose Grass Plant date: 10/15/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.74	0.00	0.00	2,210.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	14.74	0.00	0.00	2,210.42
Anticipated crop nutrient removal	270.00	18.00	18.00	0.00
Actual crop nutrient removal	84.20	8.99	96.52	415.35
Nutrient balance	-69.47	-8.99	-96.52	1,795.06
Applied to removed ratio	0.18	0.00	0.00	5.32

Fresh water applied	
29,136,717.00	*gallons*
1,073.01	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure - 2022

Sample and source description: Separator Manure - 2022

Sample date: 05/11/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	2,900.00	11,400.00							0.00
DL	100.00	100.00	100.00							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 04/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,300.00	6,000.00	14,500.00	17,300.00	8,900.00	4,300.00	6,000.00	6,000.00		25.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 09/06/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,700.00	17,400.00							0.00
DL	100.00	100.00	100.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 12/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	2,700.00	8,300.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022

Sample and source description: Lagoon Q4 - 2022

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	726.00	464.00	0.00	1.90	68.60	1,370.00								13,400.00	7,450
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	655.00	457.00	0.00	1.40	50.00	1,050.00								11,600.00	6,050
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	521.00	328.00	0.00	1.00	104.00	740.00								9,300.00	5,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	765.00	411.00	0.00	1.30	125.00	1,760.00								13,400.00	7,180
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	553.00	370.00	0.00	0.20	76.90	1,170.00								11,100.00	7,220
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

C. FRESH WATER ANALYSES

Canal

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.70								328.00	200
DL		0.01	0.01								0.01	1

Domestic Well

DW - MB

Sample description: DW - MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	1.70								989.00	0
DL		0.01	0.01								0.01	1

DW - Road

Sample description: DW - Road

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	2.00								994.00	0
DL		0.01	0.01								0.01	1

Freshwater Pond

Freshwater Pond

Freshwater Pond

Sample description: Freshwater Pond

Sample date: 08/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.50								1,870.00	0
DL		0.01	0.01								0.01	1

Irrigation Well

IW - W13

Sample description: IW - W13

Sample date: 08/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.20	252.00	272.00	999.00	301.00	0.00	2,100.00	910.00	7,570.00	4,750
DL		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

IW - W5

Sample description: IW - W5

Sample date: 08/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.00	101.00	77.30	427.00	414.00	0.00	560.00	306.00	2,440.00	2,930
DL		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

Irrigation Well

IW - W7

Sample description: IW - W7

Sample date: 08/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.40	183.00	148.00	540.00	518.00	0.00	983.00	505.00	4,830.00	1,430
DL		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

D. SOIL ANALYSES

N11

N11

Sample and source description: N11

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			7.00				
DL			0.01				

W1

W1

Sample and source description: W1

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			0.01				

W13

W13

W13

Sample and source description: W13

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			157.00				
DL			0.01				

W14

W14

Sample and source description: W14

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			0.01				

W15

W15

Sample and source description: W15

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			132.00				
DL			0.01				

W16

W16

W16

Sample and source description: W16

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			161.00				
DL			0.01				

W2

W2

Sample and source description: W2

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			43.00				
DL			0.01				

W23

W23

Sample and source description: W23

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.00				
DL			0.01				

W5

W5

W5

Sample and source description: W5

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			45.00				
DL			0.01				

W7

W7

Sample and source description: W7

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			0.01				

E. PLANT TISSUE ANALYSES

N10 - 11/14/2022: Alfalfa, hay

Field N10

Sample and source description: Field N10

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,900.00	3,200.00	24,700.00		10.00
DL	100.00	100.00	100.00		1.00

N11 - 10/06/2022: Oats, silage-soft dough

N11 - 10/06/2022: Oats, silage-soft dough

Field N11

Sample and source description: Field N11

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	1,700.00	13,100.00		7.60
DL	100.00	100.00	100.00		1.00

N11 - 06/30/2023: Corn, silage

Field N11

Sample and source description: Field N11

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	1,400.00	8,800.00		5.10
DL	100.00	100.00	100.00		1.00

N2 - 10/13/2018: Alfalfa, hay

Field N2

Sample and source description: Field N2

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,200.00	2,600.00	23,800.00		10.70
DL	100.00	100.00	100.00		1.00

N3 - 10/11/2022: Oats, silage-soft dough

Field N3

Sample and source description: Field N3

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	2,800.00	24,600.00		8.80
DL	100.00	100.00	100.00		1.00

N4 (Picardo) - 10/15/2021: Alfalfa, hay

Field N4

Sample and source description: Field N4

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	39,900.00	3,300.00	22,000.00		10.40
DL	100.00	100.00	100.00		1.00

N5 - 11/15/2020: Alfalfa, hay

Field N5

Sample and source description: Field N5

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	43,400.00	3,900.00	25,200.00		10.00
DL	100.00	100.00	100.00		1.00

N6 - 11/15/2020: Alfalfa, hay

Field N6

Sample and source description: Field N6

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	43,400.00	3,900.00	25,200.00		10.00
DL	100.00	100.00	100.00		1.00

N7 - 11/15/2020: Alfalfa, hay

Field N7

Sample and source description: Field N7

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	44,100.00	4,000.00	23,500.00		10.80
DL	100.00	100.00	100.00		1.00

N8 - 11/15/2020: Alfalfa, hay

Field N8

Sample and source description: Field N8

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	44,100.00	4,000.00	23,500.00		10.80
DL	100.00	100.00	100.00		1.00

N9 - 11/14/2022: Alfalfa, hay

Field N9

Sample and source description: Field N9

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,900.00	3,200.00	24,700.00		10.00
DL	100.00	100.00	100.00		1.00

W1 - 10/24/2022: Oats, silage-soft dough

Field W1

Sample and source description: Field W1

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,600.00	19,400.00		8.80
DL	100.00	100.00	100.00		1.00

W1 - 05/28/2023: Corn, silage

Field W1

Sample and source description: Field W1

Sample date: 09/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	2,600.00	9,900.00		5.00
DL	100.00	100.00	100.00		1.00

W13 - 10/08/2022: Oats, silage-soft dough

Field W13

Sample and source description: Field W13

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	2,200.00	17,100.00		8.30
DL	100.00	100.00	100.00		1.00

W13 - 05/28/2023: Corn, silage

Field W13

Sample and source description: Field W13

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,400.00	8,800.00		4.70
DL	100.00	100.00	100.00		1.00

W14 - 10/08/2022: Oats, silage-soft dough

Field W14

Sample and source description: Field W14

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	2,200.00	17,100.00		8.30
DL	100.00	100.00	100.00		1.00

W14 - 06/28/2023: Sudangrass, hay

Field W14

Sample and source description: Field W14

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	3,600.00	30,700.00		14.10
DL	0.01	0.01	0.01		1.00

W15 - 10/10/2022: Oats, silage-soft dough

Field W15

Sample and source description: Field W15

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	3,100.00	25,200.00		8.70
DL	100.00	100.00	100.00		1.00

W15 - 06/19/2023: Sudangrass, hay

Field W15

Sample and source description: Field W15

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	3,800.00	33,100.00		14.10
DL	100.00	100.00	100.00		1.00

W16 - 10/10/2022: Oats, silage-soft dough

Field W16

Sample and source description: Field W16

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	3,100.00	25,200.00		8.70
DL	100.00	100.00	100.00		1.00

W16 - 06/19/2023: Sudangrass, hay

Field W16

Sample and source description: Field W16

Sample date: 10/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,000.00	4,000.00	30,200.00		14.90
DL	100.00	100.00	100.00		1.00

W2 - 10/06/2022: Oats, silage-soft dough

Field W2

Sample and source description: Field W2

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	4,100.00	19,900.00		9.00
DL	100.00	100.00	100.00		1.00

W2 - 05/28/2023: Corn, silage

Field W2

Sample and source description: Field W2

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	3,200.00	9,900.00		5.70
DL	100.00	100.00	100.00		1.00

W20 - 10/10/2022: Oats, silage-soft dough

Field W20

Sample and source description: Field W20

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	2,600.00	21,900.00		9.10
DL	100.00	100.00	100.00		1.00

W23 - 10/11/2022: Oats, silage-soft dough

Field W23

Sample and source description: Field W23

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	3,200.00	28,800.00		9.80
DL	100.00	100.00	100.00		1.00

W23 - 05/22/2023: Corn, silage

Field W23

Sample and source description: Field W23

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,300.00	8,500.00		5.00
DL	100.00	100.00	100.00		1.00

W3 - 11/15/2020: Alfalfa, hay

Field W3

Sample and source description: Field W3

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	42,000.00	3,800.00	24,300.00		8.80
DL	100.00	100.00	100.00		1.00

W4 - 11/15/2020: Alfalfa, hay

Field W4

Sample and source description: Field W4

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,500.00	2,500.00	26,400.00		7.30
DL	100.00	100.00	100.00		1.00

W5 - 10/06/2022: Oats, silage-soft dough

Field W5

Sample and source description: Field W5

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,700.00	19,000.00		9.10
DL	100.00	100.00	100.00		1.00

W5 - 05/24/2023: Corn, silage

Field W5

Sample and source description: Field W5

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,900.00	9,900.00		5.90
DL	100.00	100.00	100.00		1.00

W6 - 10/05/2022: Alfalfa, hay

Field W6

Sample and source description: Field W6

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,400.00	2,800.00	24,000.00		10.60
DL	100.00	100.00	100.00		1.00

W7 - 10/07/2022: Oats, silage-soft dough

Field W7

Sample and source description: Field W7

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,700.00	3,500.00	24,600.00		9.10
DL	100.00	100.00	100.00		1.00

W7 - 05/25/2023: Corn, silage

Field W7

Sample and source description: Field W7

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	2,200.00	10,100.00		5.40
DL	100.00	100.00	100.00		1.00

W8 - 10/15/2021: Alfalfa, hay

Field W8

Sample and source description: Field W8

Sample date: 11/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	41,100.00	3,300.00	24,600.00		9.70
DL	100.00	100.00	100.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

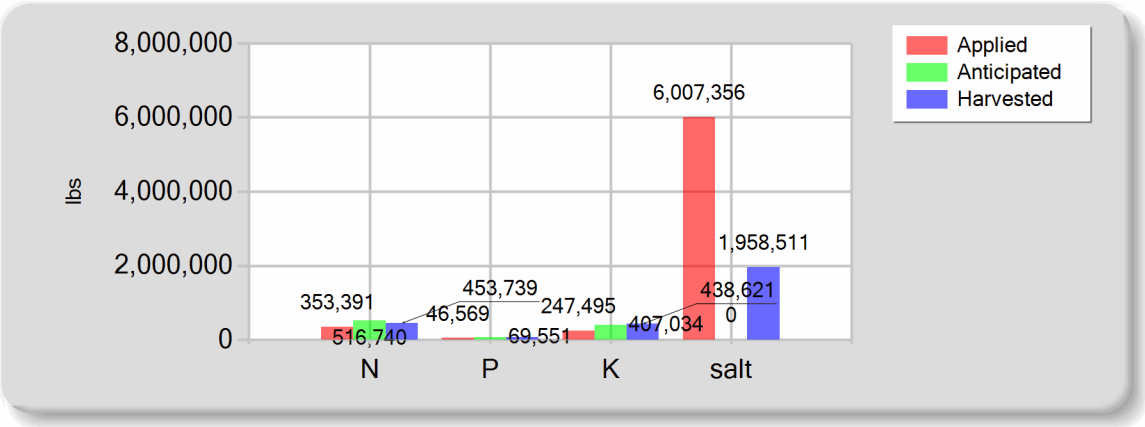
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

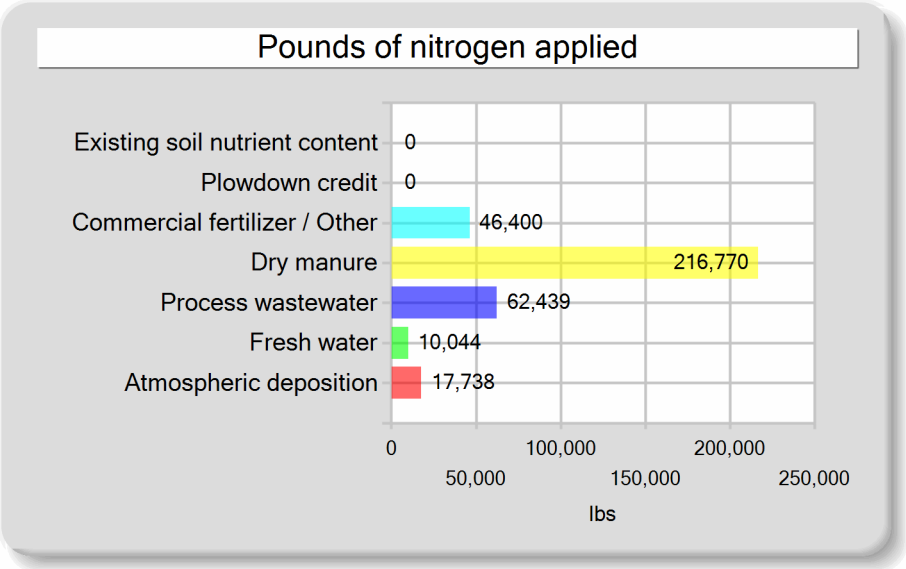
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

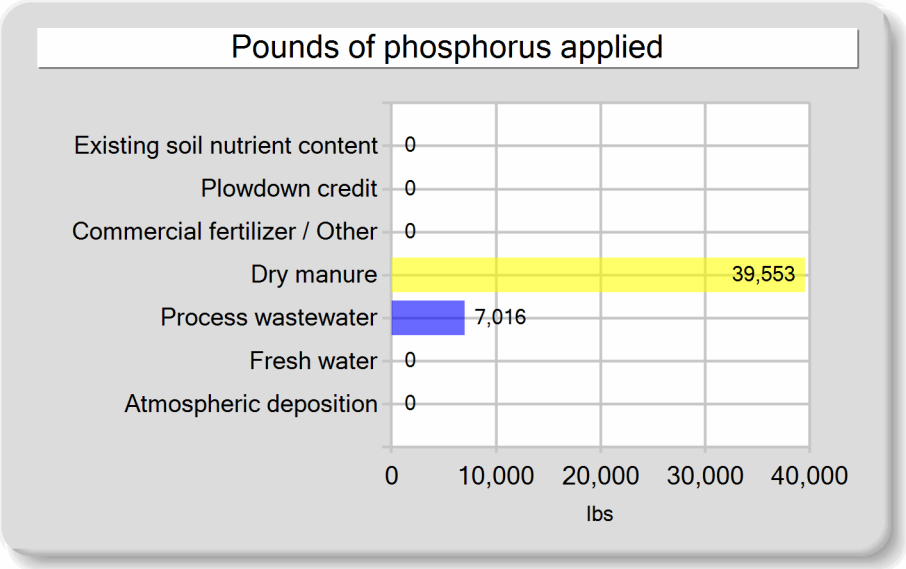
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46,400.00	0.00	0.00	0.00
Dry manure	216,770.12	39,552.90	132,498.90	580,179.60
Process wastewater	62,438.60	7,016.11	114,995.77	609,446.03
Fresh water	10,043.94	0.00	0.00	4,817,730.13
Atmospheric deposition	17,738.00	0.00	0.00	0.00
Total nutrients applied	353,390.66	46,569.01	247,494.67	6,007,355.76
Anticipated crop nutrient removal	516,740.00	69,550.60	407,034.00	0.00
Actual crop nutrient removal	453,738.62	67,317.80	438,621.10	1,958,510.57
Nutrient balance	-100,347.96	-20,748.79	-191,126.43	4,048,845.19
Applied to removed ratio	0.78	0.69	0.56	3.07

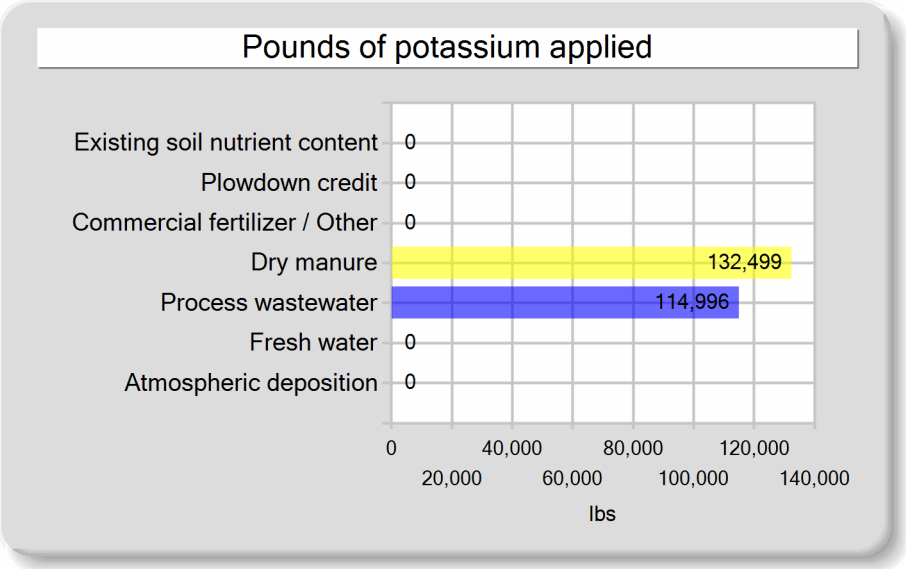
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

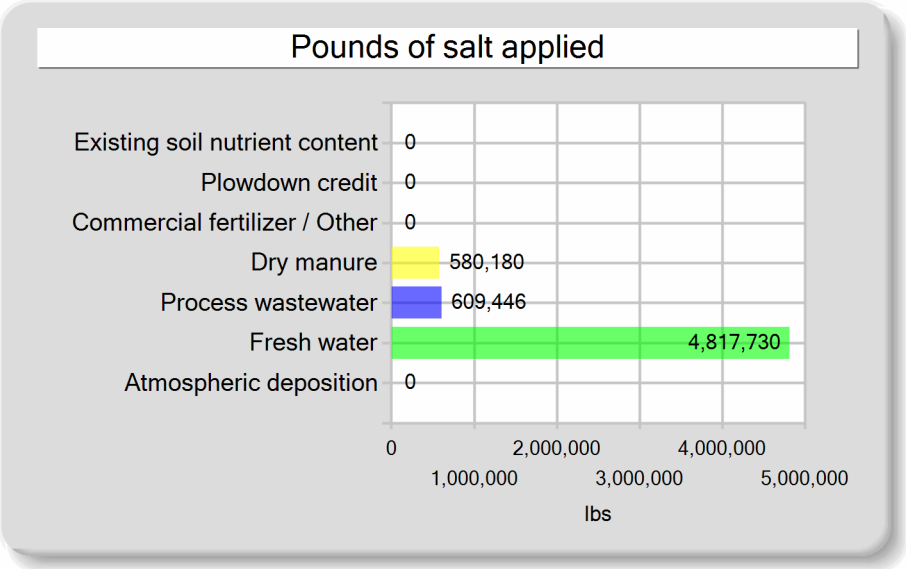


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No summer crop planted on fields N3 and W20

ANNUAL REPORT VALIDATION INFORMATION

A. VALIDATION ERRORS

The following sections contain validation errors and should be reviewed before submitting the Annual Report :

1. Harvest Events

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Jose Nunes	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jose & Odette Nunes	Robert Silveira
PRINT OR TYPE NAME	PRINT OR TYPE NAME
7-1-24	7-1-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos CA 93635
00-0021356 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23H1163
Sampled By: M. Pedroso
Requested By: Jose & Odete Nunes
Submitted Date: 08/10/23
Reported Date: 08/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TDS mg/L	CO_3 as $CaCO_3$ mg/L	HCO_3 as $CaCO_3$ mg/L	Cl mg/L	SO_4 mg/L	Ca mg/L	Mg mg/L	Na mg/L	pH at 25°C unit
1	IW-W4 (Bottle says "W5")	8/10/23	12:08	2440	2.44	ND	*		2930	ND	414	306	560	101	77.3	427	8.0
2	IW-W7	8/10/23	12:10	4830	4.83	0.4	*		1430	ND	518	505	983	183	148	540	7.9
3	IW-W13	8/10/23	12:20	7570	7.57	0.2	*		4750	ND	301	910	2100	252	272	999	7.9

***** = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos CA 93635
00-0021356 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23H1163
Sampled By: M. Pedroso
Requested By: Jose & Odete Nunes
Submitted Date: 08/10/23
Reported Date: 08/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	IW-W4 (Bottle says "W5")	8/10/23	12:08	**2440**	**2.44**	ND	0.00	0.00
2	IW-W7	8/10/23	12:10	**4830**	**4.83**	0.4		
3	IW-W13	8/10/23	12:20	**7570**	**7.57**	0.2		

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos, CA 93635

Account# 00-0021356
Account Manager: Ben Nydam
Submitted By: Jose & Odete Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:58

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1011-01	DW - MB	Well Water	M Dekeyrel		12/13/2023 14:16
23L1011-02	DW - Road	Well Water	M Dekeyrel		12/13/2023 14:17

Default Cooler Temperature on Receipt °C: 4.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos, CA 93635

Account# 00-0021356
Account Manager: Ben Nydam
Submitted By: Jose & Odete Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:58

Sample Results

Sample: DW - MB
23L1011-01 (Water)

Sampled: 12/13/2023 14:16
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.99**	mmhos/cm	0.01	1		12/19/23 11:53	SM 2510 B		BEL0706
Electrical Conductivity umhos	**989**	umhos/cm	10.0	1		12/19/23 11:53	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:16	Field		BEL0642
Nitrate Nitrogen as NO3N	**1.7**	mg/L	0.1	1	10	12/18/23 20:03	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:53	SM 4500-H+	H	BEL0706
pH	**8.0**	units	1.0	1		12/19/23 11:53	SM 4500-H+	H	BEL0706

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LABORATORY INC

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11132 Wilson Road
Los Banos, CA 93635

Account# 00-0021356
Account Manager: Ben Nydam
Submitted By: Jose & Odete Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:58

Sample Results
(Continued)

Sample: DW - Road
23L1011-02 (Water)

Sampled: 12/13/2023 14:17
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.99**	mmhos/cm	0.01	1		12/19/23 11:55	SM 2510 B		BEL0706
Electrical Conductivity umhos	**994**	umhos/cm	10.0	1		12/19/23 11:55	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:17	Field		BEL0642
Nitrate Nitrogen as NO3N	**2.0**	mg/L	0.1	1	10	12/18/23 23:02	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:55	SM 4500-H+	H	BEL0706
pH	**7.9**	units	1.0	1		12/19/23 11:55	SM 4500-H+	H	BEL0706

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DELLAVALLE™ LABORATORY INC

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos, CA 93635

Account# 00-0021356
Account Manager: Ben Nydam
Submitted By: Jose & Odete Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:58

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0686										
Blank (BEL0686-BLK1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0686-BS1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0686-BS2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEL0686-DUP1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L		5.1			0.974	10
Duplicate (BEL0686-DUP2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L		5.3			0.530	10
Matrix Spike (BEL0686-MS1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.1	106	90-110		
Matrix Spike (BEL0686-MS2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.3	102	90-110		
Reference (BEL0686-SRM1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0686-SRM2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0686-SRM3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos, CA 93635

Account# 00-0021356
Account Manager: Ben Nydam
Submitted By: Jose & Odete Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:58

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706										
Blank (BEL0706-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0706-BLK2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.9		1.0	units						
Blank (BEL0706-BLK3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.8		1.0	units						
Duplicate (BEL0706-DUP1)		**Source: 23L1017-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1.74		0.01	mmhos/cm		1.73			0.403	10
pH	7.7		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	1740		10.0	umhos/cm		1730			0.403	10
Duplicate (BEL0706-DUP2)		**Source: 23L1088-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	0.52		0.01	mmhos/cm		0.52			0.116	10
Electrical Conductivity umhos	517		10.0	umhos/cm		517			0.116	10
pH	6.7		1.0	units		6.8			0.297	10
Reference (BEL0706-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	440			umhos/cm	426.0		103	90-110		
Reference (BEL0706-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0706-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0706-SRM4)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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DELLAVALLE™
LABORATORY INC

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos, CA 93635

Account# 00-0021356
Account Manager: Ben Nydam
Submitted By: Jose & Odete Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:58

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706 (Continued)										
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM6)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM7)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM8)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L1011

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: Acct # 21356 Cons # 08 WWS 12/14/23

Results Need By

No. Samples: 2 No of Bottles:

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Jose Nunes Attn Tripe J Creek Dairy
Address: 11132 Wilson Road
City: Los Banos State: CA Zip: 93635
Telephone: Fax:
Cell/Email:
COPY TO: Cardoso Ag Services
REQUESTED BY: Jose + Odete Nunes
PROJECT: 209-617-7508
CROP:

[X] Copy of Chain [X] QA/QC Documents
Sampled By: M Dekeyrel

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

(✓) DWW1: EC, NO_3-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW-MB	12/13	2:16	4.1	Ø
2	DW-Road	12/13	2:17	5.2	Ø
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeyrel	CAS Inc	12/13/23 2:18pm	12/14/23
Second	JLF	DLV	12-14-23 12:00	12-14-23 2:NS
Third				
Fourth		DLV	12/14 14:2	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping

$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

12/14/23 14:20 23L1011

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos CA 93635
00-0021356 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1011
Sampled By: M Dekeyrel
Requested By: Jose & Odete Nunes
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/13/23	14:16	989	0.99	1.7	ND		8.0
2	DW - Road	12/13/23	14:17	994	0.99	2.0	ND		7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Jose Nunes, attn. Triple J Creek Dairy
11132 Wilson Road
Los Banos CA 93635
00-0021356 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1011
Sampled By: M Dekeyrel
Requested By: Jose & Odete Nunes
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/13/23	14:16	**989**	**0.99**	1.7	4.64	0.39
2	DW - Road	12/13/23	14:17	**994**	**0.99**	**2.0**	5.46	0.46

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.